

Minimum Standing Trust: Optimizing Zero Trust Principles Across Regulated Enterprise SaaS Security Operations

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Abstract

Regulated SaaS providers must defend exposed web applications and APIs, distributed cloud control paths, BYOD-heavy workforces, and privileged internal workflows while simultaneously preserving availability, auditability, and user experience. Zero Trust Architecture (ZTA) is widely adopted as a target model for these environments; yet practical implementations frequently devolve into either an unworkable internal deny-all posture that fractures operations, or an insufficiently rigorous configuration that leaves unjustified standing trust intact. This paper proposes the Minimum Standing Trust (MST) framework: an operations optimization model for security operations centers (SOCs) that support SaaS platforms processing regulated data, including PHI, PII, PCI-DSS-scoped data, and GDPR-governed information. MST formalizes a two-plane enforcement architecture. At the application edge, the prescribed posture is strict minimization of allowed ingress, enforced through WAF controls, NIDS visibility, bidirectional abuse-list auto-blocking (inbound and outbound, at both network and endpoint layers), and tightly governed exception handling. Inside the distributed enterprise, ZTA is implemented not as universal blocking but as the progressive removal of standing trust through strong identity, device posture, session-aware authorization, just-in-time privilege elevation, and behavioral context. The paper further argues that the operational hinge of this model is a unified telemetry plane spanning SIEM, EDR, XDR, WAF, NIDS, IAM, VPN or ZTNA, cloud, and data-access events. On this foundation, agentic AI can safely accelerate triage and response—but only when autonomy is bounded by policy, reversibility, and blast-radius constraints derived from organizational risk tolerance. The resulting framework reduces analyst burden, limits unnecessary access, improves audit-evidence generation, and offers a concrete path for aligning Zero Trust theory with cyber defense practice in regulated SaaS environments.

Keywords Minimum Standing Trust; Zero Trust Architecture; telemetry fusion; identity-centric access; agentic AI governance; reversible response; regulated SaaS; virtual security operations center

Abbreviations BYOD, CISA, DoD, EDR, GDPR, IAM, JIT, MFA, MST, NIDS, NIST, PCI-DSS, PHI, PII, SaaS, SIEM, SOC, vSOC, WAF, XDR, ZTA, ZTNA, ZTX

Introduction

NIST Special Publication 800-207 defines Zero Trust as a shift away from static, network-based perimeters toward controls centered on users, assets, and resources, and explicitly frames the model as a necessary response to remote users, BYOD programs, and cloud-hosted assets Rose et al. (2020). That description maps precisely to the modern regulated SaaS provider, and can be further generalized to any organization that supports both an externally facing application infrastructure (web or otherwise), alongside an internally facing network infrastructure used by those individuals and systems that are inside organizational boundaries—i.e. the two fundamental planes in our model. In this type of environment, it is safe to assume that the externally facing application infrastructure

lives in the Cloud;¹ identities and sessions traverse both public and private networks; support, engineering, and operations work are distributed across geographies and devices; and a constrained but consequential subset of internal roles still requires carefully governed access to data or administrative functions that affect regulated records. In this setting the protected resource is not a building, a subnet, or a conventional inside network; it is the intersection of application paths, identities, sessions, data stores, and administrative

¹ The application need not be hosted by a Cloud or PaaS provider; its main characteristics simply being derived from the fact that it is Internet-accessible and as such is subject to certain categories and types of threats that are specific to the protocols and applications in question.

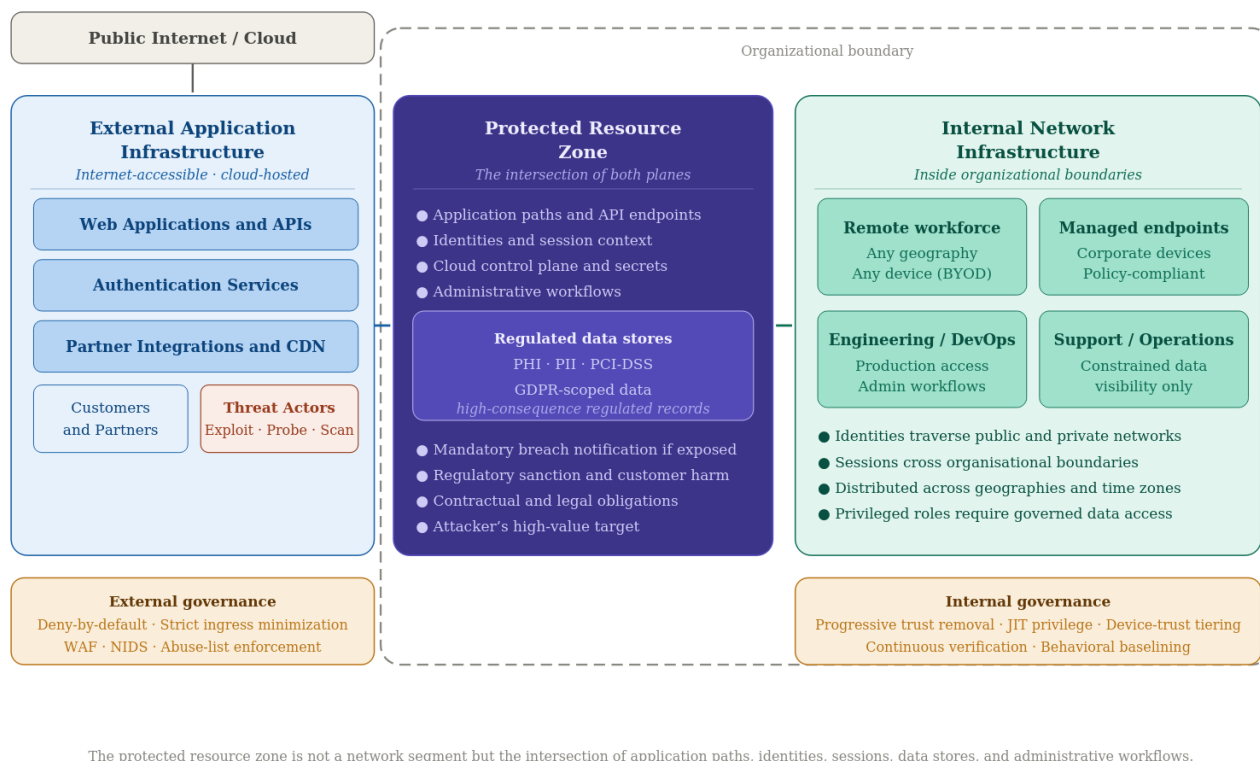


Figure 1 The two-plane operating environment. The external application infrastructure (left)—internet-accessible, cloud-hosted, and reachable by customers, partners, and threat actors—is governed by a strict ingress-minimization policy. The internal network infrastructure (right) serves the distributed organizational workforce across geographies, devices, and roles, and is governed by progressive trust removal and continuous verification. Both planes converge on a protected resource zone (centre) that is not a network segment but the intersection of application paths, identities, sessions, regulated data stores, and administrative workflows. The organizational boundary (dashed) encloses the protected resource zone and the internal plane, while the external plane is deliberately exposed to the public Internet.

workflows that collectively process or expose PHI, PII, PCI-DSS-relevant data, GDPR-scoped data, and similarly high-consequence, or otherwise regulated, information.

Figure 1 illustrates the environment that motivates the MST framework. The key structural insight is that the protected resource zone is defined by the intersection of both planes, not by a network segment or perimeter line. Anything that can reach the protected resource—whether from the external application path or from an internal administrative workflow—is subject to the same fundamental challenge: how much trust should be extended to this identity, on this device, in this session, for this operation, at this moment? The two planes differ not in the answer to that question but in the default posture, the traffic characteristics, and the governance tools most appropriate for enforcing it.

The Zero Trust principle traces its formal expression to The foundational Forrester Research report Kindervag (2010), which argued that traditional perimeter-based security architectures had become conceptually obsolete: the “trusted inside” assumption was already empirically false for enterprises with mobile workers, contractor populations, and cloud-resident data. The BeyondCorp initiative at Google operationalized this intuition at scale by moving

access controls from the network perimeter to the individual device and user, making network location irrelevant to access decisions Ward and Beyer (2014); Osborn et al. (2016). Gilman and Barth subsequently codified the design principles in a form accessible to a wider practitioner and academic audience Gilman and Barth (2017). NIST publications on Zero Trust Rose et al. (2020); Chandramouli (2023) elevated these ideas into normative federal guidance, while CISA’s Zero Trust Maturity Model CISA (2023) provided a staged implementation roadmap. The United States Department of Defense Zero Trust Strategy DoD (2022) and Executive Order 14028 on Improving the Nation’s Cybersecurity Executive Office (2021) further institutionalized Zero Trust adoption as a matter of national security policy.

Despite this rich lineage, many Zero Trust programs become operationally brittle because they are translated into a simplistic network slogan: deny everything, trust nothing, then carve back exceptions until the business can function. Security leaders face relentless external probing, credential abuse, endpoint compromise, supply-chain dependencies, and the reality that a single misrouted administrative path can expose the same data that the organization is contractually and legally obligated to protect. Yet a literal internal

deny-all posture produces predictable pathologies: exception sprawl, analyst fatigue, fragile change windows, shadow IT proliferation, and policy bypasses by teams that must keep the service running Buck et al. (2021). An enterprise can therefore accumulate more controls and more friction while achieving *less* measurable assurance.

Meanwhile, the adversary landscape facing SaaS operators has intensified. The Verizon Data Breach Investigations Report, the CrowdStrike Global Threat Report, and the Cisco Talos Year in Review consistently converge on the same leading breach vectors: credential theft, identity-based attacks, and web application exploitation Verizon (2025); CrowdStrike (2025); Cisco Talos (2025). CrowdStrike reports that identity-based initial access techniques—including the abuse of valid credentials and adversary-in-the-middle phishing—featured in the majority of intrusions it investigated in 2024, while Cisco Talos identifies valid account abuse as the single most common initial access technique for the same period CrowdStrike (2025); Cisco Talos (2025). The MITRE ATT&CK knowledge base catalogs hundreds of adversarial techniques that exploit exactly the trust assumptions Zero Trust is designed to eliminate: lateral movement through standing credentials, privilege escalation via misconfigured roles, and data exfiltration over legitimately provisioned paths Strom et al. (2018). In regulated SaaS environments, these techniques carry amplified consequences because the attacker's prize is not merely intellectual property but records that trigger mandatory breach notification, regulatory sanction, and customer harm.

This paper advances a specific, operational interpretation of Zero Trust: the goal is not universal denial, but *minimum standing trust consistent with business-required operations*—abbreviated throughout as Minimum Standing Trust (MST). The argument rests on four claims that form the spine of the paper.

First, Zero Trust fails in practice not from insufficient controls but from a category error: practitioners confuse Zero Trust with universal denial, then carve back exceptions until the business can function. The correct objective is not to maximize blocking but to eliminate *standing* trust—privileges, sessions, and access paths that persist beyond the transaction that justified them.

Second, the enforcement logic at the Internet-facing application edge must be fundamentally different from the logic governing the internal enterprise. Edge controls belong as close to deny-by-default as the service can tolerate; internal controls should progressively remove standing trust through identity, device posture, session context, and data-aware authorization. WAF, NIDS, and bidirectional reputation-based auto-blocking (inbound and outbound, at the network perimeter and at the endpoint) are edge and perimeter *tools*, not a strategy. The real message is: constrain exposure, constrain privilege, and prove continuously. Network location is not a trust signal. Standing privilege is the primary attack surface.

Third, a unified telemetry plane is the operational hinge of the entire model. Without it, every other control degenerates into isolated noise. With it, the security operations center can answer the decisive questions—who acted, from where, against which resource, with what trust context, and with what consequence—without manual event correlation.

Fourth, agentic AI can safely accelerate triage and response, but only when its autonomy is bounded by policy, reversibility, and blast-radius constraints that the organization has explicitly authorized. AI should operate inside the MST model, not alongside it as a separate problem domain. The governance criteria that determine whether

an access action is permitted are the same criteria that determine whether an automated response action is permissible.

The contribution of this paper is therefore an operational framework that bridges architectural Zero Trust guidance and modern cyber defense operations for regulated SaaS enterprises. The framework synthesizes NIST Zero Trust guidance Rose et al. (2020); Chandramouli (2023), the CISA maturity model CISA (2023), current AI governance and secure deployment guidance Tabassi et al. (2023); Autio et al. (2024); NSA et al. (2024, 2025); NCSC (2023); OWASP (2025), and the Zero Trust eXtended (ZTX) ecosystem framework Cunningham (2018). The result is not a vendor-specific product design; it is a practical operating model instantiable across different tooling environments.

The remainder of the paper is structured as follows. Section 2 situates the MST framework within the existing Zero Trust research literature. Section 3 presents the formal definition and optimization structure of the MST model. Section 4 describes the two-plane operating architecture. Sections 5 and 6 detail application-edge enforcement and internal enterprise control, respectively. Sections 7 and 8 address unified telemetry and policy-governed autonomous response. Section 9 presents a results and discussion analysis. Section 10 concludes with implications for research and practice.

Related Work

Zero Trust: Foundations and Frameworks

The intellectual lineage of Zero Trust begins with the observation that enterprise security models grounded in perimeter trust were failing empirically long before they were declared obsolete theoretically. The foundational principles of computer system design Saltzer and Schroeder (1975) articulated the *principle of least privilege* as a design requirement: every program and user should operate with the minimum set of privileges necessary to complete the job. This principle, formalized in 1975, is the earliest formal ancestor of Zero Trust. The Jericho Forum's work in the mid-2000s on “de-perimeterization” built on similar intuitions, arguing that the enterprise boundary was already dissolving and that security controls needed to migrate toward the data and identity layers. The 2010 Forrester report Kindervag (2010) coined the “Zero Trust” label and articulated the first systematic architecture for building on that insight.

The BeyondCorp initiative Ward and Beyer (2014); Osborn et al. (2016) provided the first large-scale empirical evidence that a Zero Trust model could operate at production quality without perimeter dependency. BeyondCorp shifted Google's enterprise model to device- and user-centric access controls, effectively treating the corporate network as untrusted by default. The project's public documentation established the practical engineering vocabulary that later researchers and standards bodies would inherit: device inventory, certificate-based access control, fine-grained per-application access proxies, and continuous device and user validation Beyer et al. (2017).

The Zero Trust eXtended (ZTX) ecosystem framework Cunningham (2018), developed at Forrester, extended Kindervag's architecture by explicitly naming the seven pillars of a mature Zero Trust program—network, data, workloads, devices, people, visibility and analytics, and automation and orchestration—and offering measurable maturity criteria for each. ZTX is directly referenced in the CISA maturity model CISA (2023) and is the principal scaffolding

on which the present paper's multi-pillar analysis is built. The Zero Trust Networks book Gilman and Barth (2017) synthesized Google's operational experience and prior academic work into a coherent engineering guide. The book's formulation of trust as a continuous, context-sensitive score rather than a binary network-location attribute anticipates the adaptive authorization mechanisms that later NIST guidance would formalize.

NIST and Policy Normalization

NIST SP 800-207 Rose et al. (2020) represented the maturation of Zero Trust from practitioner framework to normative federal policy. The publication defined the logical components of a Zero Trust Architecture—policy engine, policy administrator, policy enforcement point—and described seven tenets that any Zero Trust program should exhibit. These tenets are foundational to the present work. NIST SP 800-207A Chandramouli (2023) extended this work into cloud-native and multi-cloud environments by adding application-level service identity, ingress and egress gateway policy, and workload-centric enforcement to the ZTA framework.

The CISA Zero Trust Maturity Model CISA (2023) operationalized these tenets into five implementation pillars (identity, devices, networks, applications and workloads, and data) with three cross-cutting capabilities (visibility and analytics, automation and orchestration, and governance). The model's staged maturity levels (Traditional, Advanced, Optimal) provide a progression path that is compatible with the tiered trust-reduction approach advocated by the MST framework. The DOD Zero Trust Strategy DoD (2022) extended the NIST and CISA frameworks into a classified operational context, emphasizing data-layer access controls and activity-based monitoring, both of which are central to the present paper's recommendations for regulated SaaS environments.

Zero Trust in Cloud and SaaS Environments

The application of Zero Trust to cloud-native and SaaS architectures raises distinct challenges not fully addressed by perimeter-era frameworks. A 2022 comprehensive survey Syed et al. (2022) surveyed the Zero Trust literature and identified identity and access management, device health validation, and micro-segmentation as the three control categories with the most consistent empirical support across cloud deployments. The authors noted that implementation heterogeneity—different cloud providers, identity systems, and application stacks—was the principal barrier to realizing theoretical Zero Trust guarantees in production. This finding directly motivates the unified telemetry plane emphasized by the MST framework.

Early SaaS security research De Chaves et al. (2011) examined the security risks inherent in multi-tenant SaaS architectures, identifying API exposure, credential sharing, and insufficient tenant isolation as the primary attack surfaces. While predating formal ZTA guidance, their analysis correctly anticipated the application-edge threat model that later Zero Trust frameworks would address through WAF integration, API gateway policy, and service identity. A systematic review of BYOD challenges Balasubramanian et al. (2021) analyzed the specific challenges of BYOD governance in regulated industries, finding that device-trust tiering—assigning different access scopes based on device management state—was more operationally viable than binary managed/unmanaged policies. This empirical finding

is directly instantiated in the MST framework's device-aware trust tiering principle.

A multivocal literature review Buck et al. (2021) multivocal literature review found that while Zero Trust is broadly endorsed as an architectural goal, evidence of successful end-to-end implementation in enterprise environments remains sparse, and the gap between policy intent and operational deployment is a recurring theme. The MST framework explicitly addresses this gap by treating the optimization of operational throughput as a first-class design objective alongside security assurance.

Identity, Behavioral Analytics, and Adaptive Authorization

The shift from network-location trust to identity-centric trust has produced a substantial body of research on the mechanisms of adaptive authorization. The role-based access control (RBAC) model Sandhu and Samarati (1994) remains the most widely deployed authorization substrate in enterprise systems. However, the static assignment of roles conflicts with Zero Trust's requirement for continuous, context-sensitive evaluation. Attribute-based access control (ABAC) and policy-based access control (PBAC) extend RBAC by incorporating environmental attributes—device posture, session context, location, behavioral signals—into authorization decisions, moving closer to the dynamic trust evaluation that ZTA prescribes Hu et al. (2014).

Behavioral User and Entity Analytics (UEBA) provides a complementary layer. Statistical modeling of user activity—peer-group deviation analysis, temporal pattern recognition, data-access volume monitoring—can surface anomalies that evade rule-based controls Tuor et al. (2017). The MST framework incorporates behavioral context not as a substitute for identity and policy but as an evidence layer that modulates authorization confidence during a session, consistent with the continuous re-evaluation tenet of SP 800-207.

Security Policy as Optimization

The idea of formulating network and enterprise security policy as a constrained optimization problem has a growing body of prior work that directly informs the MST formalization. Early approaches to formalizing security policy optimization addressed firewall configuration, showing that achieving diverse, resilient rule sets while preserving permitted reachability is a non-trivial combinatorial problem Liu and Gouda (2008). Subnet-based network hardening problems—deciding which vulnerabilities to patch or which paths to segment to reduce attack-graph reach—are known to be NP-hard, and attack graph literature has explored approximation and heuristic approaches to making them tractable Shandilya and Nagar (2014); Garey and Johnson (1979). These contributions establish that the optimization framing of security policy is not merely metaphorical but computationally precise, and that efficient approximations exist for practical problem scales.

A parallel and older tradition concerns *attack surface minimization* as an explicit engineering objective. Howard, Pincus, and Wing Howard et al. (2005) introduced the attack surface as a measurable property of a system—the subset of its resources reachable and usable by attackers—and argued that reducing this surface should be a first-order design objective. Manadhata and Wing Manadhata and Wing (2011) subsequently formalized this as a quantitative metric based on entry points, exit points, and channel resources, and established

it as a peer-reviewed evaluation tool for comparing relative security postures across system versions and configurations. This attack surface tradition connects directly to the MST formulation: the permitted action set $\mathcal{T}$ is the attack surface a successful attacker can exploit, and the trust surplus $\mathcal{T} \setminus \mathcal{L}$ is precisely the avoidable portion of that surface. MST generalises attack surface minimization by introducing the operational-cost constraint that prevents over-minimization.

The present paper builds on both traditions. First, it extends the action space from network flows and firewall rules to the richer tuple space of (identity, device, session, resource, operation) that characterizes modern SaaS access patterns. Second, it introduces the action-space-time graph as a unified representation that captures both spatial access structure and temporal verification dynamics in a single model amenable to graph-theoretic analysis, and connects Zero Trust policy to the weighted minimum subgraph cover problem (Appendix A). Third, it integrates reputation-based pre-filtering—operationalized through feeds such as AbuseIPDB AbuseIPDB (2024) and Spamhaus Spamhaus Project (2024)—as a mechanism that reduces the effective action space $\mathcal{S}$ before the optimization problem is formulated, further tightening the achievable minimum.

AI and Automation in Security Operations

The integration of machine learning and AI into security operations has accelerated substantially in the period 2023–2025. Alert fatigue—the tendency of high-volume detection platforms to produce more events than analysts can meaningfully review—has been identified as one of the most consequential operational problems facing modern SOC's Verizon (2025). AI-assisted alert enrichment and triage has shown measurable efficiency gains in controlled evaluations, particularly for known-indicator matching and correlated-event summarization Husari et al. (2017). However, the extension of AI autonomy into active response introduces new failure modes. Adversarial manipulation of detection models Papernot et al. (2016), distribution shift between training and operational data, and the amplification of erroneous automation decisions at machine speed are all recognized risks.

Recent work on large language model (LLM)-assisted security operations has further sharpened these concerns. A recent survey Ferrag et al. (2025) survey LLM applications across threat intelligence, vulnerability analysis, and autonomous penetration testing, finding that while LLMs substantially accelerate analyst workflows, they introduce new attack surfaces including prompt injection against agentic security tools and hallucination-induced false negatives in threat reasoning. A 2024 survey of LLM applications in cybersecurity Motlagh et al. (2024) examine the specific risks of deploying autonomous LLM agents in SOC environments, arguing that action boundaries and reversibility constraints are necessary preconditions for safe deployment—a finding that directly corroborates the bounded-autonomy ladder formalized in Section 8.

NIST's AI Risk Management Framework Tabassi et al. (2023) and its Generative AI Profile Autio et al. (2024) provide a governance vocabulary for managing these risks. The MST framework's bounded-autonomy principle—restricting autonomous action to high-confidence, low-blast-radius, reversible decisions—is directly derived from the trustworthiness and oversight principles of the AI RMF. OWASP's agentic application security guidance OWASP (2025) extends these principles into the specific context of tool-using

AI agents deployed within security workflows, addressing prompt injection, excessive tool use, and unsupervised policy modification as primary threat vectors against agentic SecOps tooling. A taxonomy of language model risks Weidinger et al. (2022) provide a taxonomy of sociotechnical harms from autonomous AI agents that is applicable to the agentic SecOps context, identifying irreversibility and scope creep as the highest-consequence risk dimensions—consistent with the blast-radius criteria used in the MST autonomy ladder.

Synthesis and Gap Identification

The extant literature collectively establishes that Zero Trust is architecturally sound, empirically motivated, and increasingly normative. The persistent gap is between architectural specification and operational deployment. The MST framework addresses this gap by: (1) formalizing the optimization objective as a multi-criteria problem with explicit tradeoffs between security assurance, operational throughput, and audit evidence quality; (2) differentiating enforcement posture by exposure plane rather than applying symmetric policy; and (3) integrating bounded AI autonomy into the operations model as a governed capability rather than an aspirational feature. No prior work known to the authors has combined these three contributions into a unified operational framework specifically calibrated to regulated SaaS environments.

The Minimum Standing Trust Model

The MST framework treats Zero Trust implementation as a constrained optimization problem rather than a policy posture. The objective is to find the minimum-cost permitted action set $\mathcal{T}^*$ that covers all business-required operations $\mathcal{L}$ while continuously verifying that each permitted action meets a resource-sensitive evidence threshold $V(a_i, t) \geq \theta(\rho_i, t)$. Each action is weighted by a risk-exposure cost c_i (the cost of incorrectly permitting it) and a denial cost d_i (the operational cost of incorrectly blocking it). The result is a two-term objective that makes explicit the tradeoff that most Zero Trust programs resolve implicitly and badly: minimizing unnecessary trust while preserving operational completeness.

The full mathematical formulation—including binary integer programming variables, weighted objective function, action-space-time graph representation, complexity analysis, and solution approaches—is presented in Appendix A. The operational sections that follow are self-contained and do not require familiarity with the formal model, but readers working on algorithmic policy enforcement or platform implementation will find the appendix provides the necessary precision.

The MST model has three direct operational implications that shape the architecture described in Sections 4–8. First, the verification score $V(a_i, t)$ can only be computed accurately if telemetry from all relevant planes is fused—making unified telemetry a correctness requirement, not a convenience. Second, the risk weight c_i and the threshold $\theta(\rho_i)$ differ fundamentally between the exposed application edge and the internal enterprise, which is why symmetric control policy across both planes is structurally wrong. Third, the blast-radius and reversibility criteria used to bound AI autonomous response are derived from the same c_i and θ parameterization: high-consequence resources require human approval before their access policy changes for exactly the same reason they require higher verification thresholds.

A Two-Plane Operating Architecture

The most useful way to operationalize the MST model is to separate the environment into two enforcement planes. The first plane is the *external application edge*: the exposed SaaS surface where north-south traffic reaches web applications, APIs, authentication services, and narrowly scoped administrative interfaces. The second plane is the *internal distributed enterprise*: the workforce, endpoints, service accounts, cloud control paths, engineering tools, support workflows, and data-adjacent systems that sustain the service behind the scenes. These two planes are structurally related but are not governed well by the same default action.

The separation of a defended network into an exposed external zone and a controlled internal zone is not itself novel; it reflects current practice across mature enterprise security programs and is implicit in NIST SP 800-207A's ingress/egress gateway model Chandramouli (2023) and in the CISA maturity model's application and network pillars CISA (2023). The contribution of the two-plane architecture in this paper is not the zone separation itself, but its *explicit mapping to the MST optimization structure*: the action space $\mathcal{S}$, the risk weights c_i , the denial costs d_i , and the verification threshold function $\theta(\rho_i, t)$ take on fundamentally different parameter values in the two planes, which is why applying symmetric control policy across both planes is not merely suboptimal but structurally inconsistent with the optimization problem's geometry. The gap between symmetric-policy programs and zone-differentiated programs is consistently documented in Zero Trust implementation reviews Buck et al. (2021).

Edge traffic is comparatively easier to constrain because the set of legitimate flows is narrower, more observable, and more closely tied to the application's business purpose. Internal enterprise traffic is more heterogeneous and more tightly coupled to changing work patterns, which makes blanket deny-all policies far more likely to produce operational self-harm. NIST SP 800-207A reinforces this separation by emphasizing granular application-level policies, service identities, ingress and egress gateways, and enforcement that works across hybrid and multi-cloud environments Chandramouli (2023). CISA's maturity model further organizes implementation around identity, devices, networks, applications and workloads, and data, with visibility and analytics, automation and orchestration, and governance as cross-cutting capabilities CISA (2023).

This separation leads to a practical design rule: *control density should be highest where exposure and consequence intersect*. At the external edge, that means hardening ingress, minimizing accepted traffic, and placing the burden of justification on everything unusual. Internally, it means reducing standing privilege, continuously validating identities and devices, and using session-level and behavioral context to decide whether an action remains consistent with the role, the asset, and the data involved. The bridge between the two planes is a unified telemetry and policy layer that turns events from WAF, NIDS, SIEM, EDR, XDR, IAM, VPN or ZTNA, cloud platforms, and data-access systems into a shared operational picture rather than a collection of disconnected alarms.

Figure 2 summarizes the proposed operating model. The primary architectural mistake in many Zero Trust programs is not excessive control, but excessive *symmetry*: when the same binary control philosophy is imposed on both the exposed edge and the internal distributed enterprise, either the edge remains too open or the enterprise becomes too brittle Buck et al. (2021). The two-plane MST architecture avoids this tradeoff by allowing strict minimization

where the traffic set is narrow and progressive trust reduction where business operations remain inherently dynamic.

Threat Model

The MST framework is designed to address a specific and well-documented adversary profile relevant to regulated SaaS providers. Based on the MITRE ATT&CK framework Strom et al. (2018) and annual empirical breach intelligence Verizon (2025); CrowdStrike (2025); Cisco Talos (2025), the principal threats are:

- **External application attacks:** Web application exploitation (OWASP Top Ten vulnerabilities), API abuse, automated credential stuffing, and bot-driven enumeration. These attacks target the application edge and are best addressed by the external plane controls.
- **Identity and credential compromise:** Phishing, adversarial token theft, session hijacking, and MFA bypass techniques (including adversary-in-the-middle proxies such as Evilginx). These attacks exploit the trust implicit in valid credentials and motivate continuous session re-verification.
- **Privilege escalation and lateral movement:** Exploitation of overprivileged service accounts, standing administrative access, and misconfigured IAM roles. These attacks are addressed by just-in-time privilege, device-trust tiering, and behavioral baselining.
- **Data exfiltration:** Abuse of legitimate data-access paths to extract regulated records, often using credentials obtained in prior stages. This threat motivates data-aware access policy, field-level masking, and egress monitoring.
- **Insider threat and support-account misuse:** Authorized users acting outside normal behavioral baselines, whether through negligence, coercion, or malicious intent. This motivates behavioral analytics and just-enough privilege enforcement.
- **Command-and-control and known-bad egress:** Established footholds—whether on compromised endpoints, rogue servers, or hijacked service accounts—that attempt to communicate with adversary-controlled infrastructure for command reception, tool staging, or data exfiltration. A large fraction of this traffic resolves to IP addresses already catalogued on public and commercial reputation feeds (AbuseIPDB, Spamhaus DROP/EDROP, and vendor threat-intelligence lists). This motivates symmetric, bidirectional reputation-based blocking at both the network perimeter (NIDS/firewall) and the endpoint (EDR-enforced egress policy)—inbound connections from known-bad sources and outbound connections to known-bad destinations are both automatically denied.

The MST framework does not claim to defeat all of these threats in isolation. Rather, it provides the architectural substrate within which specific control technologies operate with maximum effectiveness and minimum operational friction.

Application-Edge Enforcement

At the SaaS edge, Zero Trust should be strict. The exposed application surface should be reduced to the smallest practical set of domains, protocols, ports, paths, and partner integrations required for customer use and legitimate business operations. If traffic is not necessary for the service to function, it should not reach the service. Administrative interfaces should not be broadly reachable from the

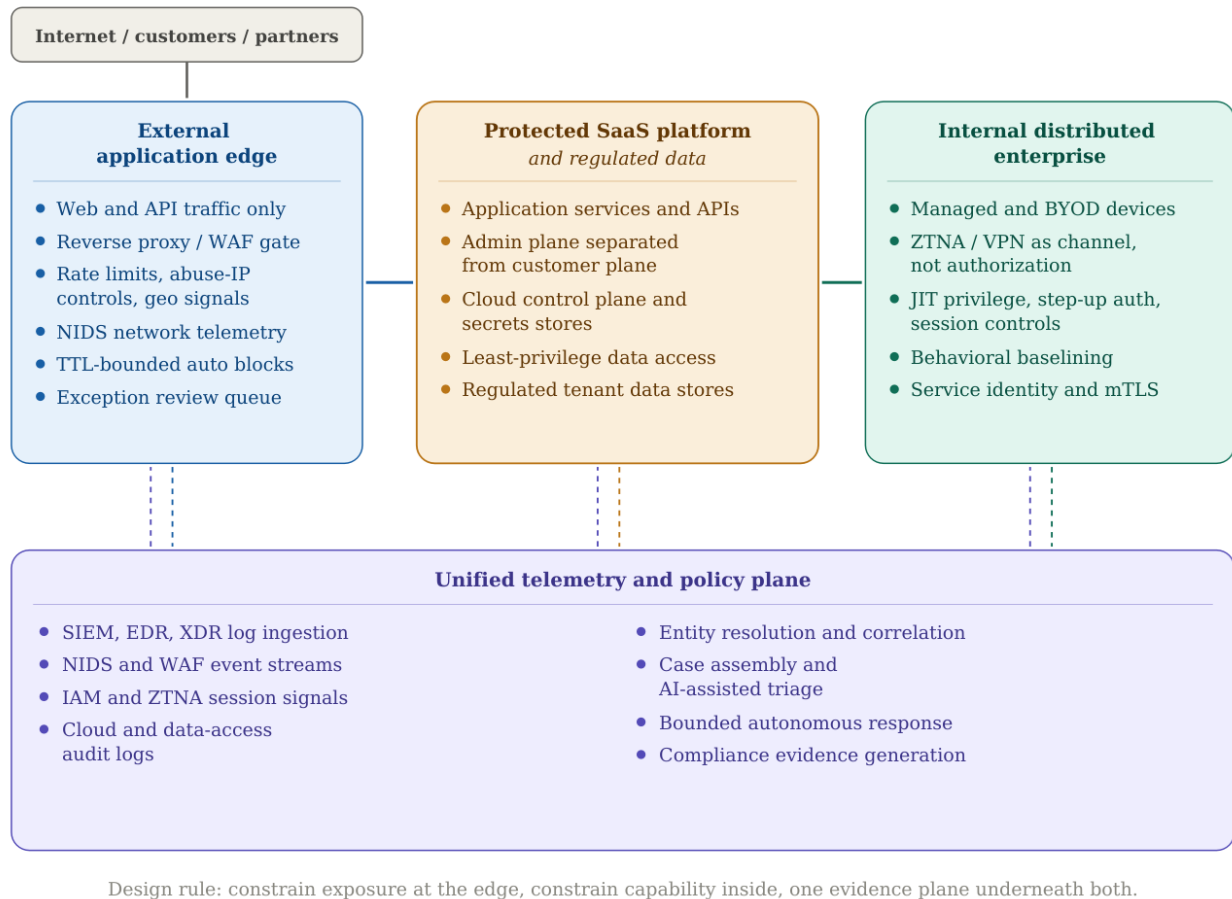


Figure 2 Two-plane operating model for regulated SaaS Zero Trust. The external plane applies deny-by-default pressure at the SaaS edge through tightly constrained ingress, application-aware controls, and documented exception handling. The internal plane progressively removes standing trust through identity, device, session, and data-aware control. Both planes depend on a unified telemetry and policy layer that supports consistent vSOC decision-making.

public Internet; customer APIs should be versioned and documented; unpublished or legacy endpoints should be retired rather than merely hidden; and edge policy should clearly distinguish among customer traffic, machine-to-machine traffic, partner integrations, and administrative control paths.

SP 800-207A is especially relevant here because it emphasizes application and service identities in addition to network parameters, enabling a SaaS provider to control not only who is calling an interface but which service is calling which other service and under what policy Chandramouli (2023). This is critical in microservice architectures where east-west service-to-service traffic within the platform can itself represent an attack surface if lateral movement is possible through a compromised service identity.

In practice, the edge control stack typically includes a reverse proxy or gateway, a WAF, abuse and reputation intelligence, rate-limiting controls, and one or more NIDS vantage points that can observe north-south traffic and selected east-west paths behind the edge. These controls are not interchangeable. A WAF is strongest when it

enforces application-aware policy around malformed requests, exploit patterns, bot behavior, API misuse, and route-specific protections. A NIDS is strongest when it identifies suspicious network behavior, scanning activity, protocol anomalies, lateral movement indicators, or traffic patterns that become meaningful only when viewed in sequence. Using WAF for behavioral pattern detection or NIDS for application-layer enforcement produces both functional gaps and telemetry noise; the architectural separation of responsibilities matters.

Because the legitimate edge surface is relatively compact, autonomous prevention is most realistic at this plane. High-confidence controls can be allowed to act by default when their scope is narrow and their rollback path is simple. The most important such control is *bidirectional auto-blocking of IP addresses appearing on curated reputation feeds*. Sources listed on feeds such as AbuseIPDB AbuseIPDB (2024), the Spamhaus DROP and EDROP lists Spamhaus Project (2024), and vendor-provided threat-intelligence bundles represent, by construction, hosts that

have demonstrated adversarial activity against other organizations. Permitting connections to or from them serves no legitimate business purpose. Accordingly, the SecOps platform should ingest one or more such feeds continuously and automatically drop both inbound connections originating from listed IPs and outbound connections destined for them—the symmetric posture that is central to the MST interpretation of the edge. The symmetry matters: inbound blocking alone addresses probing, scanning, and credential-stuffing attempts, but outbound blocking is what denies command-and-control callbacks, tool-staging retrievals, and exfiltration paths from any foothold an adversary has already established. Neither direction is complete without the other. Other autonomous edge controls—short-lived blocks on repeated exploit-like request patterns, suppression of commodity scanning behavior with no plausible customer use case—complement this reputation-based foundation. Geo-based controls can be appropriate when justified by the service footprint, contractual obligations, or observed threat concentration, but they should be treated as coarse risk signals rather than proof of maliciousness. A workable program gives blocked traffic a documented exception path so that the organization can say “no by default” while preserving a controlled allowlisting mechanism for legitimate business cases.

A further edge consideration is egress discipline from sensitive workloads and administrative enclaves. Regulated SaaS providers often invest more in ingress filtering than in outbound control, even though data exfiltration, command-and-control callbacks, and accidental exposure can ride legitimate-looking outbound paths. Zero Trust at the edge therefore benefits from a symmetrical design: published inbound paths should be explicit, but so should the outbound destinations and protocols available to systems handling regulated data or administrative control. Service-to-service traffic, update channels, partner APIs, and observability exports should be enumerated and monitored. Everything else should at minimum be measurable and, where feasible, denied by policy.

External Zero Trust also carries a compliance dividend. Edge decisions are easier to explain to auditors and customers when they are tied to documented policy, constrained exposure, and loggable enforcement actions. The same evidence that helps a vSOC explain why a source was blocked can help a compliance or privacy team explain how the service minimizes unnecessary access to regulated data. A disciplined edge therefore simultaneously reduces attack surface and improves defensibility during audits, customer reviews, and breach-response inquiries.

Internal Enterprise Control Without Operational Self-Harm

The internal enterprise is where Zero Trust programs most frequently lose credibility. The workforce is distributed. Some users are on managed corporate endpoints; some are on contractor machines; and some are on BYOD devices that may be acceptable for productivity workflows but inappropriate for sensitive administrative access. Engineers need to push code, observe production behavior, and solve incidents under time pressure. Support staff may require constrained visibility into customer-impacting cases. Security administrators need enough reach to investigate and contain activity. These realities do not invalidate Zero Trust, but they require a more precise implementation model than simple deny-all.

The ZTX framework’s seven pillars Cunningham (2018) provide a useful organizing structure for the internal enterprise controls advocated by the MST framework. The following subsections address the four pillars most directly implicated in SecOps operations for regulated SaaS: identity, devices, data, and behavioral analytics. The remaining pillars—network, workloads, and automation—are addressed in Sections 7 and 8.

Identity as the Primary Policy Object

NIST’s Zero Trust guidance removes implicit trust based on network location and instead places authentication and authorization before each session to an enterprise resource Rose et al. (2020). For regulated SaaS operations, this means strong identity proofing, multifactor authentication, and preferably phishing-resistant mechanisms (e.g., FIDO2/WebAuthn passkeys) for privileged users; role- and attribute-aware authorization tied to concrete job function; short session lifetimes for sensitive access; and policy checks that re-evaluate context during the session rather than only at logon.

The internal question is not simply whether a user reached the corporate network. It is whether this identity, on this device, in this session, for this task, should continue to retain this level of access. This formulation is consistent with the ABAC and PBAC models identified in the research literature Hu et al. (2014) as the access control paradigms most compatible with Zero Trust’s dynamic authorization requirements. Operationally, it means that authorization decisions are not static grants managed at provisioning time but continuous evaluations driven by current context.

A related concern is service account and machine identity. In regulated SaaS environments, service accounts frequently hold elevated permissions that human accounts do not, because they were provisioned early in the platform’s development under fewer controls. Machine-to-machine authentication via short-lived certificates or OIDC token federation, combined with just-in-time (JIT) privilege elevation for sensitive operations, should be applied to service identities with the same rigor as to human identities.

Device-Aware Trust Tiering

BYOD does not need to be eliminated to make Zero Trust credible, but it does require honest segmentation of what those devices are permitted to do Balasubramanian et al. (2021). A mature SaaS program can allow low-sensitivity collaboration and basic ticketing from lightly trusted devices while reserving administrative consoles, production shells, sensitive data paths, and broad incident-response capabilities for managed and policy-compliant endpoints. Device posture checks can include verified operating-system patch state, endpoint protection agent health and version, full-disk encryption status, certificate presence, screen-lock policy compliance, and absence of jailbreak or unauthorized administrative tool installation.

The practical effect is that a user may be fully authenticated while the device remains insufficiently trusted for a high-consequence action. This tiering avoids the binary choice between “block all unmanaged devices” and “allow everything.” It also provides a defensible audit position: the organization can demonstrate that regulated data paths are accessible only from devices meeting documented posture standards.

Just-in-Time and Just-Enough Privilege

Persistent administrative access is one of the most durable forms of unjustified trust in modern enterprises. A regulated SaaS provider should minimize standing access to production systems, data stores, cloud control planes, and security controls. Elevation should be time-bounded, reason-coded, and linked to a ticket, incident, or approved maintenance window. For sensitive data operations, field-level masking, tokenization, or explicit approval workflows are often more appropriate than broad record-level access.

This is where Zero Trust and privacy engineering overlap operationally: the most effective way to protect regulated data is frequently to avoid exposing it in the first place. The principle of least privilege Saltzer and Schroeder (1975) requires not just that the initial access grant be minimal, but that the grant be regularly reviewed and revoked when the business justification lapses. Automated JIT systems that provision and expire privilege on verified request—integrating with change management ticketing, on-call rotation schedules, and data classification policies—operationalize this principle at the scale and speed that a SaaS engineering organization requires.

Behavioral Context and Continuous Verification

Behavioral modeling is compatible with but not identical to Zero Trust. It helps detect when a nominally authorized identity is acting outside its expected pattern, providing evidence that enriches rather than replaces policy-based decisions. Internal behavioral baselining can be constructed around role, time of day, application usage patterns, administrative history, data-access volume, source device, network path, and peer-group norms Tuor et al. (2017). These signals raise or lower response confidence. Used properly, they function as contextual evidence layered onto identity and policy; used poorly, they generate opaque noise that degrades analyst trust in the detection platform.

The key is to treat behavioral deviation as a modulator of the verification score $V(a)$ in the MST model—a signal that triggers re-verification, step-up authentication, or session scope narrowing—rather than as an independent enforcement mechanism. This integration is well-supported by the UEBA research literature, which consistently finds that behavioral signals have the highest operational value when combined with identity and policy context Tuor et al. (2017).

Endpoint-Enforced Reputation Blocking

The reputation-based auto-blocking posture described in Section 5 for the application edge has a direct symmetric counterpart inside the internal enterprise, enforced at the endpoint rather than at the network perimeter. EDR and XDR agents deployed on managed endpoints are already positioned to observe every outbound connection from every process on every host. By integrating the same curated reputation feeds that drive edge-level blocking—AbuseIPDB, Spamhaus, vendor threat intelligence AbuseIPDB (2024); Spamhaus Project (2024)—into the EDR policy engine, the platform can automatically block any process on any managed endpoint from initiating an outbound connection to a known-bad IP, and from accepting an inbound connection from one. The NIST SP 800-94 guide to intrusion detection and prevention systems establishes reputation-list filtering as a standard IDS/IPS capability Scarfone

and Mell (2007); applying it at the endpoint rather than only at the network perimeter closes the visibility gap created by work-from-home patterns, where endpoint traffic often bypasses the corporate network boundary entirely.

This symmetric posture is structurally important. An adversary who has established a foothold on an internal endpoint (via phishing, supply-chain compromise, or any other initial access vector) will typically attempt to reach adversary-controlled infrastructure for command reception, lateral tool retrieval, or data staging. If the destination infrastructure is on a public reputation feed—and a substantial fraction of active adversary infrastructure is, because reputation feeds are populated in near-real-time by honeypot networks, abuse reports, and passive DNS collectors—then endpoint-enforced egress blocking denies the callback before the adversary can execute subsequent stages. Combined with the edge-level blocking of inbound connections from the same lists, the organization denies both the initial probe and the subsequent exfiltration channel with a single, operationally cheap control that requires no per-decision human judgment.

The control is well-suited to the bounded-autonomy framework (Section 8): the signal quality is high (the IP is on a curated feed with explicit evidence of malicious activity), the blast radius is low (a single connection is denied; a business-legitimate connection would rarely target a reputation-listed IP), and the action is fully reversible (the IP can be whitelisted in policy if a false positive occurs). These properties place reputation blocking firmly within the region of the MST model that admits full autonomous execution.

A useful shorthand emerges from these five principles: internally, Zero Trust is best implemented as the *progressive removal of standing trust* rather than as the elimination of all connectivity. An organization matures when fewer people retain persistent privilege, fewer unmanaged devices can touch sensitive paths, fewer sessions continue without renewed proof, fewer anomalous actions go unexplained, and fewer endpoints can reach known-bad infrastructure. This is a realistic and sustainable maturity trajectory for distributed SaaS operations, and one that accommodates the mobility and device heterogeneity of modern knowledge work.

Unified Telemetry and Decisioning

Architecture becomes operations only when telemetry is fused. Many enterprises already operate capable point products: SIEM for aggregation and correlation, EDR or XDR for endpoint and cross-domain detection, WAF for application protection, NIDS for network visibility, identity services for authentication and conditional access, cloud consoles for infrastructure telemetry, and data-access logs for sensitive operations. The problem is not typically missing tools. The problem is that each tool tries to tell its own story about what happened, while the analyst is left to merge those partial narratives manually, under time pressure, against an adversary who is not similarly constrained.

This operational fragmentation is well-documented. Alert fatigue—the state in which detection volume exceeds the analyst’s capacity for meaningful triage—is among the most cited productivity barriers in SOC research Verizon (2025). The consequence is not merely wasted analyst time; it is systematic under-investigation of genuine signals buried in commodity noise, which directly degrades detection and containment performance.

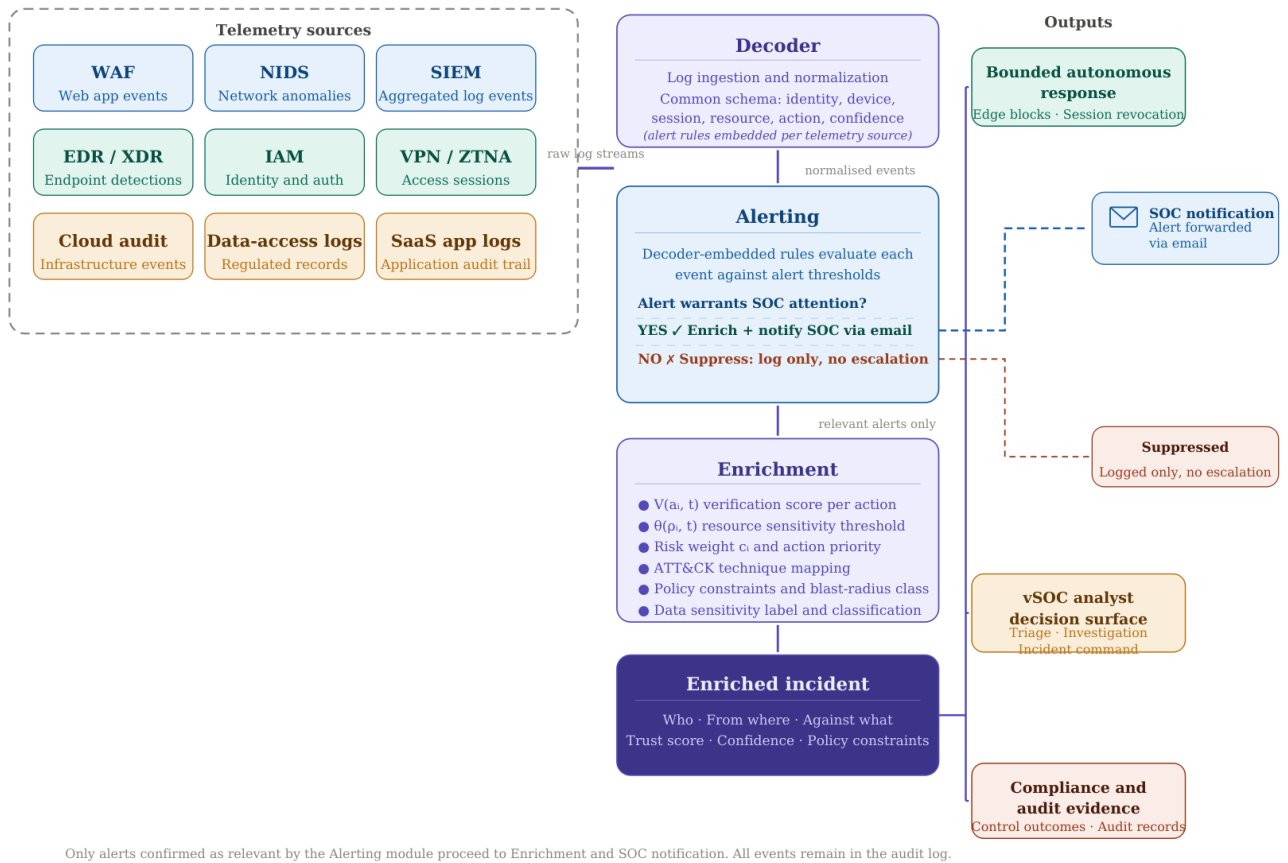


Figure 3 Unified Telemetry & Data Ingestion Pipeline. All nine telemetry sources—WAF, NIDS, SIEM, EDR/XDR, IAM, VPN/ZTNA, cloud audit, data-access logs, and SaaS application logs—are grouped within a single collection boundary and feed a *Decoder* module that performs log ingestion and normalization into a common schema. Decoder-embedded alert rules then drive an *Alerting* module that determines whether each normalized event warrants SOC attention: events that pass are forwarded to the SOC via email notification and proceed to *Enrichment*; events that do not pass are suppressed and retained in the audit log only. The enrichment layer computes the verification score $V(a_i, t)$, applies the resource threshold $\theta(\rho_i, t)$, assigns risk weight c_i , maps adversary techniques, and attaches policy constraints. The resulting enriched incident object is routed to three consumers: bounded autonomous response, the vSOC analyst decision surface, and compliance and audit evidence generation.

A practical SecOps design therefore requires a single ingestion and normalization pipeline. The data model should revolve around a small set of stable objects: identity, device, session, resource, action, evidence quality, and business consequence. WAF events, NIDS alerts, EDR detections, cloud audit logs, VPN or ZTNA sessions, SaaS application logs, and data-access events should be normalized into a common schema so the vSOC can answer a small set of decisive questions quickly: who or what acted; from where; against which resource; using which trust context; with what likely consequence; and with what confidence. Without that normalization, automated response will either stay too conservative to matter or act on incomplete evidence with unpredictable outcomes.

Figure 3 illustrates the four-stage pipeline. The dashed boundary on the left groups all nine telemetry sources, making explicit that the architectural problem being solved is not missing tools but the absence of a single ingestion path. The *Decoder* stage handles log normalization and carries the alert rules that are specific to

each telemetry source—the logic that knows, for example, what a NIDS rule-fire threshold or an EDR behavioral indicator threshold should look like before it merits analyst attention. The *Alerting* stage applies those rules as a filter gate: only events that cross the relevance threshold proceed to enrichment and SOC notification; all others are logged but not escalated, directly addressing alert fatigue. The *Enrichment* stage then computes the MST formal model variables ($V(a_i, t)$, $\theta(\rho_i, t)$, and c_i) and attaches ATT&CK mappings and policy constraints. This architecture ensures that the analyst receives an enriched incident rather than a raw fragment, and that computational resources are spent only on alerts the decoder has already qualified as relevant.

Evidence Classification and Decision Value

To support sound decisions, the normalization layer should classify evidence on two dimensions: *technical trustworthiness* (how reliably

does this signal indicate adversarial activity?) and *operational decision value* (how much does this signal affect the response decision for this resource?). Some events are strong technical indicators of maliciousness but weak indicators of business consequence. Others are technically weak but operationally consequential because they involve a privileged identity or a regulated dataset. A mature pipeline preserves both dimensions.

For example, an EDR alert on a kiosk device may be technically high-confidence but operationally low-consequence. A weaker behavioral anomaly tied to a production administrator who has just opened a sensitive data-access path from an unmanaged endpoint may be technically uncertain but operationally critical. The unified telemetry layer resolves this asymmetry by enriching events with resource classification, identity role, session context, and data-sensitivity labels before routing them to the response decision engine.

Compliance as an Operational Dividend

A unified telemetry plane is also where compliance becomes operational rather than retrospective. In regulated SaaS firms, security and compliance have historically operated in separate evidence loops: one loop for incident handling and another for audits, attestations, and customer questionnaires. That separation introduces cost and delay because analysts resolve the same event that auditors and compliance teams later need re-explained from a different analytical frame. A shared data plane can eliminate this duplication. Case notes, control outcomes, policy decisions, and exception records created during security operations should also be available as evidence for governance, privacy, and assurance workflows, with appropriate access controls and retention policies.

This integration supports what the NIST Cybersecurity Framework 2.0 (NIST (2024)) identifies as the “Govern” function: the organizational structures, policies, processes, and oversight mechanisms that determine how cybersecurity risk decisions are made, prioritized, and communicated. Unified telemetry makes that governance function continuous rather than periodic, enabling real-time status rather than point-in-time snapshots.

The design implication bears repeating: unifying telemetry at the dashboard layer improves visualization; unifying it at the decision layer improves operations.²

The vSOC should receive *incidents*, not fragments. A suspicious event should arrive with identity context, device trust tier, resource sensitivity, prior related signals, probable adversary technique mapping where applicable, and any policy constraints on what automation is permitted to do next.

Policy-Governed Autonomous Response

The rise of agentic AI has revived an old security ambition: a SOC that can not only observe and recommend, but act at machine speed. That ambition is operationally sound only if action is bounded

by policy rather than by model confidence alone—and *the policy in question is the MST model itself*. AI governance in security operations is not a separate problem domain; it is an instance of the same optimization structure. The permitted action set $\mathcal{T}^*$ that governs human access decisions also governs what an AI agent may do autonomously: if an action would require human authorization when a human requests it, it requires human authorization when an AI agent recommends or executes it.

The governance literature establishes the requirements clearly. NIST’s AI Risk Management Framework and its Generative AI Profile emphasize that AI systems must be governed as a risk-management discipline, with trustworthiness, oversight, measurement, and context-appropriate controls built in at design time Tabassi et al. (2023); Autio et al. (2024). The 2025 NSA/CISA/FBI joint guidance on deploying AI systems securely stresses governance, data provenance, adversarial robustness, and continuous monitoring for model drift NSA et al. (2024, 2025). OWASP’s agentic application security guidance identifies prompt injection, excessive tool grant, and unsupervised policy modification as the primary attack vectors against SOC-deployed AI agents OWASP (2025). Ferrag et al. survey LLM applications in security operations and find that while LLMs substantially accelerate analyst workflows, they introduce hallucination risk in threat reasoning and are directly susceptible to adversarial manipulation of their input telemetry Ferrag et al. (2025). Motlagh et al. argue that action boundaries and reversibility constraints are necessary preconditions for safe autonomous SOC deployment—not optional governance preferences Motlagh et al. (2024).

The integration of these requirements into the MST model produces a straightforward design rule: *AI agents in security operations should follow the same trust constraints as human analysts, applied to response actions rather than access decisions*. Concretely, an AI agent executing a response action must satisfy the same criteria as any other permitted action: the verification evidence must be sufficient ($V(a_i, t) \geq \theta$), the risk weight must be low (c_i small), and the action must be reversible and time-bounded. This means that an AI agent may block an IP address or trigger step-up authentication autonomously on the same evidence that would justify a human analyst doing so; it may not rewrite access policy, grant durable privilege, or take any action that a human analyst would need approval to take. The MST model thus provides a unified governance surface for both human and AI actors in the security operations workflow.

For cyber defense operations, the safest synthesis is *bounded autonomy*. An AI-enabled assistant provides high value when it enriches alerts, correlates related evidence, drafts concise case summaries, explains likely adversary behavior, and recommends next steps. It becomes more sensitive when allowed to suppress noisy alerts, trigger step-up authentication, temporarily isolate a host, revoke a session token, or block an IP range. It becomes high risk when it can rewrite policy broadly, grant or revoke enduring access, or take actions materially affecting customers or regulated data without immediate human review.

The Bounded-Autonomy Ladder

The practical test for whether an action belongs on the autonomous side of the ladder is not whether the model appears confident. It is whether the action is simultaneously: (1) high-confidence based

² All three authors are affiliated with Athena Security Group. Athena Core Athena Security Group (2026a) and Athena Pallas Athena Security Group (2026b) are commercially available instantiations of the unified telemetry and AI-assisted analyst patterns described in this section. They are cited only as disclosed implementation examples. The framework is intended to be vendor-neutral and is applicable across different tooling environments.

on multiple corroborating signals; (2) low-blast-radius with respect to customer-facing services and regulated data; (3) reversible within a short operational window; and (4) supported by a documented playbook that encodes the organization's explicit risk tolerance for that action class.

These criteria explain why temporary edge blocking often qualifies for autonomous execution before permanent access revocation does. A short-lived block on a source matching multiple abuse signals, triggering a known exploit rule, and affecting no preapproved partner is a more appropriate autonomous action than a long-lived access change against an internal administrator during an active production incident.

The design is also relevant to reinforcement and adaptive learning within the response engine. Statistical models are well-suited to optimizing alert-priority thresholds, ranking playbook choices, learning suppression patterns for duplicate noise, and calibrating when the cost of intervention delay exceeds the cost of a reversible action. They are poorly suited to unconstrained self-directed policy modification in production. If learning systems are used to improve autonomous action over time, the reward function must incorporate not only speed and detection gains, but also false-block cost, customer impact, action reversals, and exception volume Papernot et al. (2016). Otherwise the system will optimize the wrong objective.

Human Factors and SOC Trust in Automation

There is also a human-factors rationale for bounded autonomy that the literature on automation-induced complacency makes compelling. Analysts will extend trust to automated response only when the system behaves predictably, explains its choices, and stays within known boundaries. A platform that behaves unpredictably—jumping from alert summarization to disruptive containment actions without transparent reasoning—teaches the SOC to disable automation rather than expand it Husari et al. (2017). Agentic security tooling should therefore behave more like a disciplined junior analyst with very fast hands than like an unsupervised decision-maker. It should show its work, stay within policy rails, and leave a reviewable record.

Figures 4 and Table 1 together illustrate that the safest automation program is one that expands machine speed where reversibility is high and contracts it where policy consequences are broad. Organizations should also resist confusing a more capable model with better policy. An AI assistant can become dramatically more fluent at summarizing incidents without improving the correctness of the response actions the organization has authorized. Conversely, a conservatively bounded policy can remain safe even when the AI layer is only modestly capable, because it limits the damage of incorrect inference. This separation of model capability from policy authority is a fundamental governance principle for agentic security tooling.

Every autonomous action should be explainable in operational terms. The record should show: what evidence triggered the action; what policy authorized it; how long the action was intended to persist; who could reverse it; and whether the event touched regulated data or customer-facing systems. This is not only sound governance but sound engineering. When incidents occur, organizations need to know not just what the adversary attempted, but what the defense platform decided on the organization's behalf and on what basis.

Expected Outcomes and Argument Summary

Summary of the Central Argument

The argument of this paper can be stated plainly. Zero Trust is not a product category. Network location is not a trust signal. Standing privilege is the enemy. Compliance evidence is a byproduct of good architecture, not its purpose. These are not marketing slogans; they are operational commitments with specific engineering consequences.

The Minimum Standing Trust model operationalizes these commitments in four steps. Step one: at the Internet-facing application edge, enforce deny-by-default using WAF, NIDS, and abuse-list controls as *instruments* of that posture—not as the posture itself. Step two: inside the enterprise, eliminate standing privilege through identity-centric access, device-trust tiering, JIT elevation, and continuous session verification. The goal is that no adversary who acquires a credential can inherit the access it once carried indefinitely. Step three: fuse all telemetry into a single decision plane so that the security operations center receives incidents, not fragments, and can answer who acted, from where, against what, with what evidence. Step four: let AI operate inside this model—automating what is reversible, narrow, and pre-authorized while requiring human judgment for anything that cannot be quickly undone or that touches regulated data at scale. The compliance records that regulators and auditors need are a natural output of this architecture, not a parallel documentation effort.

Expected Operational Outcomes

A Zero Trust operations model is credible only if it produces measurable improvements in outcomes rather than merely increasing control count or coverage claims. For a regulated SaaS enterprise, the MST framework is expected to produce four principal outcome improvements.

Reduction in unjustified access. The progressive removal of standing trust, JIT privilege, and device-trust tiering collectively shrink the attack surface available to an adversary who has acquired a valid credential. This directly reduces the dwell time and lateral movement capability implied by a credential-theft event. Credential and identity-based attacks remain the dominant initial-access vector across all major annual threat intelligence reports Verizon (2025); CrowdStrike (2025); Cisco Talos (2025).

Reduction in analyst toil. Unified telemetry and bounded AI automation reduce the manual correlation work imposed on analysts by fragmented detection stacks. Alert enrichment, case summarization, and duplicate suppression are the highest-volume, lowest-value analyst tasks; automating them within the bounded-autonomy framework frees analyst capacity for high-judgment work such as threat hunting, policy refinement, and incident command.

Improvement in detection and containment speed. A shared telemetry pipeline with pre-enriched incident records reduces mean-time-to-detect and mean-time-to-contain by eliminating manual pivot time. When an analyst receives an incident rather than a fragment, the time from alert to containment decision is reduced by the elimination of the context-assembly phase.

Improvement in audit-evidence generation. Unified telemetry with compliance-tagged event records transforms audit preparation from a retrospective evidence-gathering exercise into a continuous operational byproduct. The evidence a regulator or customer questionnaire requires—access logs, control outcomes, exception

Automate what is narrow, reversible, and evidenced — require human approval where blast radius is broad or rollback is hard.

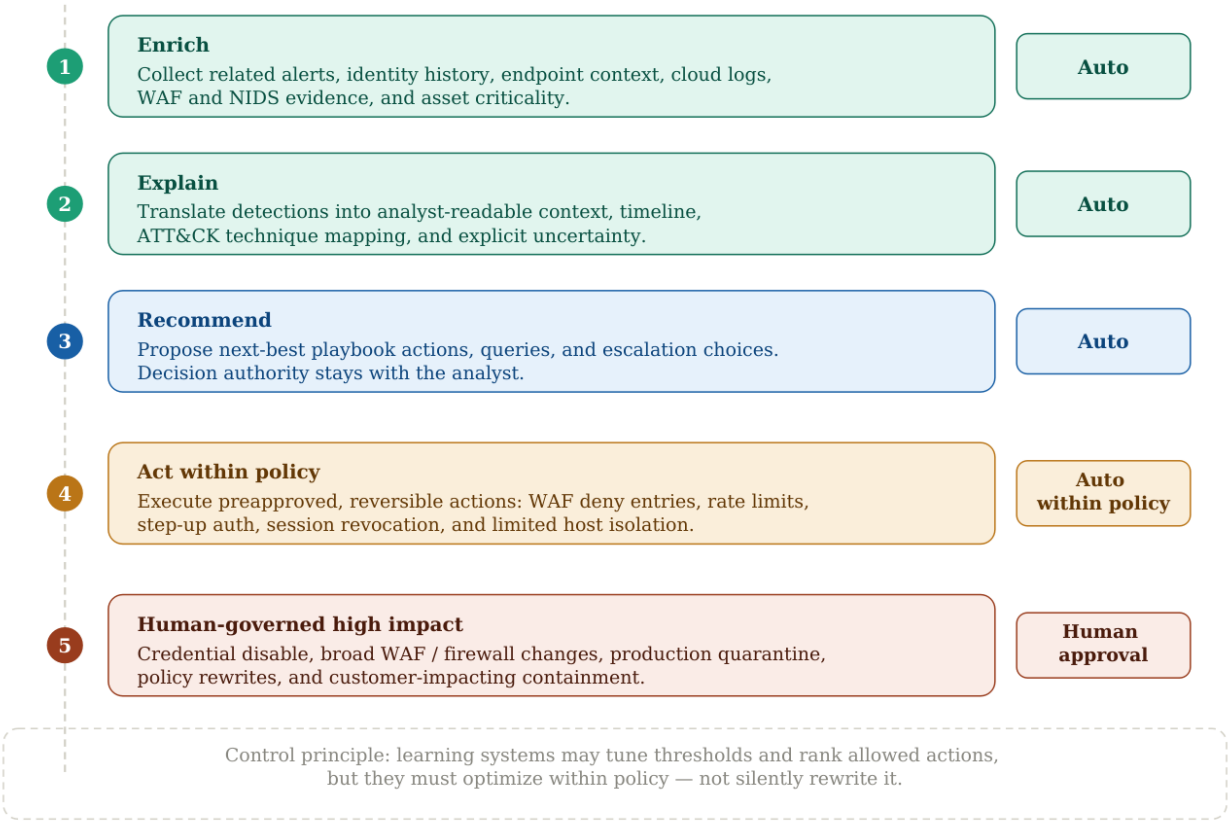


Figure 4 Bounded-autonomy ladder for the vSOC. Low-risk functions such as enrichment, correlation, summarization, and duplicate-noise suppression can be automated aggressively. As blast radius, irreversibility, or customer impact increase, autonomous action should narrow and human approval becomes mandatory.

records, policy decisions—exists as a natural output of the operational platform rather than a separately constructed artifact.

Performance Metrics and Governance Cadence

The appropriate scorecard for the MST framework balances four dimensions: security effectiveness, analyst efficiency, business friction, and evidence quality. Useful quantitative measures include: mean time to detect (MTTD); mean time to contain (MTTC); mean time to recover (MTTR); false-block rate (the fraction of automated blocks that are later reversed as erroneous); exception-turnaround time (time from exception request to documented resolution); percentage of analyst time on repeatable low-value tasks; percentage of autonomous actions subsequently reversed; and the fraction of alert volume arriving with sufficient context to support a decision without manual pivoting.

In regulated environments, a further class of metrics is operationally essential: evidence latency (how quickly can an event

be reconstructed for audit?); control-to-case traceability (can every control action be linked to a governing policy and a ticket?); and answer time for auditor or customer inquiries about specific access events.

An explicit multi-objective optimization model is useful for framing governance reviews honestly. Security teams are often informally evaluated on whether they block enough adversarial activity while formally measured on availability, audit readiness, and customer trust. These objectives can pull in opposite directions, and the temptation is to resolve the tension implicitly rather than explicitly. The MST optimization structure (Appendix A, Equation A.3) makes the tradeoff visible: reducing $|\mathcal{T} \setminus \mathcal{L}|$ is the security objective; maintaining $\mathcal{L} \subseteq \mathcal{T}$ is the availability constraint; and the verification threshold θ parameterizes the risk tolerance of the organization at each resource class. Governance reviews should track all three dimensions simultaneously.

Table 1 Response decision matrix for policy-governed autonomous response. Signal quality, blast radius, and reversibility together determine whether an action is appropriate for automatic execution or requires human review.

Event class	Typical signal quality	Blast radius / Default action	autonomous	Human-review trigger
Known-bad Internet source	High confidence from abuse feed plus exploit or scanning context	Low blast radius when block has short TTL and no partner dependency	Temporary IP block or rate limit at edge	Partner traffic, repeated exception requests, or unexpected business impact
Repeated WAF exploit attempts	High when rule precision is validated on the protected route	Low to moderate; reversible at edge controls	Short-lived block, challenge, or request throttling	Possible false positive on customer workflow or API client
Managed endpoint malware	High for confirmed malware or command-and-control indicators	Moderate; rollback available but host isolation affects user productivity	Quarantine artifact; isolate host if policy conditions are met	Executive or on-call operational asset, or uncertainty about detection quality
Duplicate low-value alerts	Medium to high if repetition pattern is stable	Very low; fully reversible	Mute or suppress with expiration and sampling review	Pattern drift, new asset class, or prior false suppression
Privileged session anomaly	Medium; behavior signal strengthened by identity and data context	Moderate to high depending on account and active task	Step-up authentication, token revocation, or narrowed session scope	Critical user, sensitive maintenance window, or customer-visible risk
Broad policy rewrite or permanent access denial	Varies; confidence alone is insufficient	High; often difficult to reverse without operational fallout	No autonomous execution	Always human approval

Validation and Red-Team Considerations

Validation must be explicit and structured. Edge policies should be exercised through red-team and purple-team scenarios simulating exploit traffic, abuse-list hits, false-attribution cases, and partner-traffic exceptions. Internal Zero Trust controls should be tested against stolen-session scenarios, unmanaged-device access attempts, privilege escalation paths, support-account misuse, and insider-style behavioral anomalies. AI-enabled playbooks should be run in shadow mode with before-and-after comparison of analyst actions, response speed, reversal rates, and business impact, before being promoted to live execution authority.

Without structured validation, organizations tend to overestimate both detection precision and readiness for autonomous action. The bounded-autonomy ladder in Figure 4 and the decision matrix in Table 1 are useful not just as design guides but as evaluation rubrics: an organization can assess whether its current autonomous action program stays within the cells that the framework designates as safe for automation.

These reviews should engage more than the SOC. In a regulated SaaS company, the consequences of Zero Trust policy are distributed across engineering, support, privacy, compliance, product, and customer trust functions. A regular cross-functional governance cadence is therefore essential. At minimum, the organization should review: the precision of automated blocks; the exception patterns introduced by geo or reputation controls; the policy drift introduced by repeated emergency overrides; the failure modes of behavioral detections; and the cases in which bounded automation either prevented harm or created avoidable friction. These reviews convert isolated lessons learned into structured policy refinement.

Limitations and Scope Boundaries

The MST framework addresses operational architecture and governance structure; it does not prescribe specific tool vendors, identity provider configurations, or regulatory compliance mappings. Organizations will need to adapt the framework’s principles to their specific regulatory context (HIPAA, PCI-DSS, GDPR, SOC 2, and others impose different specific requirements), technology stack, workforce scale, and risk tolerance. The formal model in Appendix A is intended to clarify the optimization structure rather than to be computed algorithmically in practice; the $V(a)$ scoring function and the θ thresholds are policy parameters that each organization must define based on its own risk profile and data classification standards.

The framework also does not resolve the challenge of third-party and supply-chain trust. SaaS providers depend on cloud infrastructure, open-source components, and vendor integrations that introduce trust relationships outside the direct control of the enterprise security program. While the two-plane architecture provides a structural basis for evaluating third-party access, a full supply-chain Zero Trust model is a substantial research and operational program in its own right, and is identified as a direction for future work.

Conclusion

Zero Trust is not a product category. Network location is not a trust signal. Standing privilege is the enemy. Compliance evidence is a byproduct of good architecture, not its purpose.

These four commitments are not rhetorical. Each has a specific operational consequence. Zero Trust is not a product category means that no tool purchase implements it; it is a continuous operating discipline enacted through policy, verification, and governance.

Network location is not a trust signal means that being inside the corporate network confers nothing; every access decision requires current evidence of identity, device posture, and session legitimacy. Standing privilege is the enemy means that persistent access grants—service accounts, never-expiring tokens, broadly scoped admin roles—are the attack surface that adversaries actually exploit, and eliminating them is the highest-leverage security investment an organization can make. Compliance evidence is a byproduct means that organizations that build good architecture generate the audit records regulators need as a side-effect; organizations that build for compliance and treat security as secondary get neither.

In a regulated SaaS enterprise, the Minimum Standing Trust framework operationalizes these commitments: deny-by-default at the exposed application edge; identity-, device-, and session-aware internal control with JIT privilege throughout; unified telemetry across detection and access systems; and policy-governed AI autonomous response bounded by the same trust criteria that govern human access.

The MST framework presented in this paper makes four contributions to the Zero Trust literature.

First, it reframes the Zero Trust implementation problem as a constrained optimization with competing objectives—minimizing unnecessary trust while preserving operational completeness—and shows why monotonic control accumulation fails as a strategy. This reframing explains empirically documented implementation failures that prior frameworks left without a structural account Buck et al. (2021).

Second, it prescribes asymmetric enforcement posture across the two planes and sharpens the control hierarchy: WAF, NIDS, and reputation controls are edge instruments, not strategy; identity, device posture, JIT privilege, and session verification are the substance of Zero Trust in the internal enterprise.

Third, it establishes unified telemetry as a correctness requirement of the MST model—not a convenience—and provides a four-stage pipeline architecture (Decoder, Alerting, Enrichment, Enriched Incident) that operationalizes this requirement at the decision layer rather than at the dashboard layer.

Fourth, it integrates AI governance into the MST model as a first-class concern, showing that the same trust criteria governing human access decisions govern AI autonomous response actions, and providing the bounded-autonomy ladder as a practical policy instrument for both.

A consequence of these four contributions, developed formally in Appendix A, is that the MST optimum corresponds to a principled minimum on the enterprise attack surface in the established attack-surface sense Manadhata and Wing (2011): the permitted action set $\mathcal{T}^*$ is the smallest attacker-exploitable surface consistent with the organization's business requirements. Bidirectional reputation-based blocking, applied symmetrically at both the network perimeter and the endpoint, further reduces the effective action space by removing from consideration any action whose counterparty is already known to be adversarial. The full mathematical formalization, including the weighted ILP, action-space-time graph, complexity analysis, and solution approaches, appears in Appendix A.

This operating model does not solve every problem. It does, however, close the gap between Zero Trust in theory and cyber defense in practice. It gives the vSOC a clearer decision surface. It reduces the standing privilege that adversaries can inherit through credential compromise. It creates a structured path for AI-assisted

defense without surrendering policy authority to opaque automation. And it generates the evidence a regulated SaaS provider needs not only to defend its platform, but also to explain its controls to customers, auditors, and regulators.

For organizations operating under the mobile, distributed, and data-regulated conditions that now characterize modern enterprise computing, the MST framework offers a practical and theoretically grounded path to the intersection of Zero Trust principles and effective cyber defense operations.

Future work. Several important directions remain open. The most pressing empirical question is whether the MST model's core hypothesis holds in practice: that eliminating standing trust—measured by credential exposure, dwell time, and lateral movement reach—is a reliable proxy for reducing breach impact. Longitudinal studies of Zero Trust maturity and security outcomes, building on the multivocal review methodology of Buck et al. Buck et al. (2021), would provide a rigorous empirical path to this validation. Instantiating the action-space-time graph on provenance graph datasets Hassan et al. (2020) would provide a concrete experimental substrate for evaluating the MST subgraph cover formulation at system-call granularity. Developing efficient MIP formulations calibrated to enterprise-scale action spaces—leveraging the bipartite structure of the actor-resource graph to reduce problem dimensionality—is a tractable near-term algorithmic research agenda. The extension of the two-plane architecture to supply-chain and fourth-party trust relationships introduces additional graph layers (vendor nodes, dependency edges) whose MST formulation remains open. Finally, the governance implications of agentic AI in security operations—including the formal treatment of irreversibility constraints on the x_i variables, liability for autonomous access decisions, and regulatory standards for AI-governed policy enforcement—are emerging questions at the intersection of computer science and law that the security research community is well-positioned to address.

Conflicts of interest

All three authors are affiliated with Athena Security Group. Peter J. Worth Jr. is founder, president, and CEO of Athena Security Group, and is concurrently a PhD candidate in the Department of Electrical Engineering and Computer Science at Florida Atlantic University. Dr. Ionut Cardei is a Professor in the Department of Electrical Engineering and Computer Science at Florida Atlantic University and serves as Chief Scientist at Athena Security Group. Dr. Chase Cunningham serves on the Leadership team of Athena Security Group and is the founder of the Dr. Zero Trust podcast. Athena Core and Athena Pallas are discussed in the paper only as disclosed implementation examples of the unified telemetry and AI-assisted response patterns described by the Minimum Standing Trust framework; the framework is intended to be vendor-neutral and applicable beyond any specific product or platform.

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Data availability

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Author contributions statement

P.J.W., C.C., and I.C. jointly developed the Minimum Standing Trust framework and the two-plane operating architecture. P.J.W. led the writing, the formal optimization model, and the figure and table design. C.C. contributed the Zero Trust eXtended (ZTX) ecosystem framing, the four-claim argumentative spine, and the policy-level critiques that shaped the control hierarchy and conclusions. I.C. contributed the optimization-theoretic analysis, the action-space-time graph formulation, the complexity and tractability analysis, and the attack surface minimization treatment. All three authors reviewed and approved the final manuscript.

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Formal Statement of the Minimum Standing Trust Optimization Problem

Why Zero Trust Implementations Fail as Monotonic Optimization

Before formalizing the MST model, it is worth establishing precisely why the dominant failure mode of Zero Trust programs takes the form it does. Zero Trust practitioners and academics alike have documented a recurrent pattern: programs begin with an architectural goal of radical access reduction, accumulate controls as their maturity increases, yet do not proportionally improve measurable security outcomes Buck et al. (2021). Buck et al. Buck et al. (2021) identify this gap systematically across a multivocal review of enterprise Zero Trust deployments, finding that control

accumulation without outcome measurement is the most consistently reported implementation failure mode.

This failure mode reflects a specific structural error: treating Zero Trust deployment as a *monotonic optimization*—one in which adding more controls or making existing controls stricter constitutes monotonic progress toward the security objective. Monotonic optimization is appropriate when the objective function is convex and unconstrained. Security posture is neither. The security benefit of each additional control exhibits diminishing returns once the highest-risk paths are blocked, while the operational cost of each additional control (analyst burden, exception rate, business friction) accumulates linearly. The resulting program reaches a local optimum far short of the global security objective while simultaneously overshooting the operational feasibility constraint. The organization cannot distinguish this outcome from genuine security progress because it is measuring control density rather than risk reduction.

We are therefore careful to distinguish the MST claim from a simple inversion: the goal is not to *minimize* controls, but to find the *minimum-cost trust set* that satisfies both the security and operational constraints simultaneously. This is a constrained optimization problem, not a greedy minimization.

Action Space and Decision Variables

Let $\mathcal{S} = \{a_1, a_2, \dots, a_n\}$ denote the *action space*: the finite set of all access actions possible in the enterprise at a given point in time. Each action $a_i \in \mathcal{S}$ is a tuple:

$$a_i = (\iota_i, \delta_i, \rho_i, \sigma_i, \omega_i) \quad (\text{A.1})$$

where ι_i is the authenticated identity, δ_i is the device trust tier, ρ_i is the target resource and its data classification, σ_i is the session context (freshness, MFA strength, network path), and ω_i is the requested operation (read, write, execute, elevate, etc.).

Introduce binary decision variables $x_i \in \{0, 1\}$ for each $a_i \in \mathcal{S}$:

$$x_i = \begin{cases} 1 & \text{if action } a_i \text{ is permitted (included in } \mathcal{T}) \\ 0 & \text{otherwise} \end{cases} \quad (\text{A.2})$$

The *permitted action set* is then $\mathcal{T} = \{a_i \in \mathcal{S} : x_i = 1\}$ and the *legitimate business-required set* $\mathcal{L} \subseteq \mathcal{S}$ is defined as the subset of actions for which a documented business or operational justification exists. Let $\ell_i \in \{0, 1\}$ denote membership in $\mathcal{L}$.

As a **uniform-cost baseline**, the MST problem in its simplest form seeks the minimum-cardinality permitted set that covers all legitimate actions and satisfies a verification threshold:

$$\mathcal{T}^* = \arg \min_{\mathcal{T}} |\mathcal{T} \setminus \mathcal{L}| \quad \text{subject to} \quad \mathcal{L} \subseteq \mathcal{T}, \quad \forall a_i \in \mathcal{T} : V(a_i) \geq \theta(\rho_i) \quad (\text{A.3})$$

Where:

- $c_i \geq 0$ is the **risk-exposure weight** for action a_i —the cost of incorrectly permitting it. Allowable values are any non-negative real number. In practice, organisations assign c_i on a relative scale calibrated to data classification and role sensitivity: a read-only query against a non-regulated internal dashboard might be assigned $c_i = 1$; a privileged write to a PHI record store by a service account $c_i = 50$; an administrative action on the cloud control plane by an unmanaged device $c_i = 100$ or higher. The scale itself is arbitrary—what matters is the ratio

between values, since the optimiser trades off c_i terms against d_i terms. A practical starting point is a four-tier classification: $c_i \in \{1, 10, 50, 100\}$ mapped to low, medium, high, and critical data sensitivity respectively.

- $d_i \geq 0$ is the **denial-cost weight** for action a_i —the cost of incorrectly blocking it when it is legitimately required. Allowable values are any non-negative real number. For actions that are business-critical and time-sensitive, d_i should be high: blocking an on-call engineer's read access to production logs during an incident might carry $d_i = 200$, reflecting the cost of delayed remediation. For speculative or low-frequency access patterns, d_i may be close to zero. In the uniform-cost baseline (Equation A.3), $d_i \rightarrow \infty$ for all legitimate actions, which forces the optimiser to permit every member of $\mathcal{L}$ unconditionally. In the weighted formulation, finite d_i values allow the policy designer to express that some legitimate actions carry acceptable denial risk—useful for soft-blocking patterns that are legitimate but rarely exercised, where temporary disruption is tolerable.
- $\theta(\rho_i) \in [0, 1]$ is the **verification threshold** for the target resource ρ_i —the minimum composite evidence score an action must achieve before it may even be considered for permission. Unlike c_i and d_i , which enter the objective function, θ acts as a hard gate: any action with $V(a_i) < \theta(\rho_i)$ is ineligible regardless of its cost weights. Allowable values are in $[0, 1]$, where $\theta = 0$ imposes no verification requirement and $\theta = 1$ requires perfect evidence on every dimension—an unachievable bar in practice. A workable implementation maps θ to resource sensitivity class: $\theta = 0.4$ for low-sensitivity internal resources; $\theta = 0.6$ for production systems; $\theta = 0.8$ for regulated data stores; and $\theta = 0.95$ for cloud control plane and secrets management interfaces. The verification score $V(a_i)$ that is compared against $\theta(\rho_i)$ is itself a composite of identity assurance level, device trust tier, session freshness, MFA strength, and behavioural conformance, each component scored in $[0, 1]$ and aggregated by policy-defined weighting.

Equation (A.3) is equivalent to setting $c_i = 1$ and $d_i \rightarrow \infty$ for all i in the full formulation: every unjustified permitted action costs equally, and every denied legitimate action is infinitely costly. This uniform version is operationally interpretable—minimize the count of unnecessary permissions—but it ignores the fact that permitting a production-administrator session to a regulated data store carries far greater risk than permitting a read-only dashboard query. Section A.3 generalises it accordingly.

Weighted Objective Function

Section A.2 established that in the uniform-cost baseline, every unjustified permitted action costs equally ($c_i = 1$) and every denied legitimate action is infinitely costly ($d_i \rightarrow \infty$). The full formulation replaces those fixed values with action-specific weights, producing the **MST optimization problem**:

$$\min_{\mathbf{x}} \sum_{i=1}^n c_i x_i (1 - \ell_i) + \sum_{i=1}^n d_i (1 - x_i) \ell_i \quad (\text{A.4})$$

subject to:

$$x_i = 1 \quad \forall a_i : \ell_i = 1 \quad (\text{business completeness}) \quad (\text{A.5})$$

$$x_i \leq 1[V(a_i) \geq \theta(\rho_i)] \quad \forall a_i \quad (\text{verification sufficiency}) \quad (\text{A.6})$$

$$x_i \in \{0, 1\} \quad \forall i \in \{1, \dots, n\} \quad (\text{A.7})$$

Plain-English reading of Equation (A.4). The equation asks: find the set of permission decisions—each x_i is either grant ($x_i = 1$) or deny ($x_i = 0$)—that minimises the total cost of getting those decisions wrong. It accumulates cost in exactly two ways. The first term charges c_i whenever an action that has no business justification ($\ell_i = 0$) is nevertheless permitted ($x_i = 1$): this is the price of unnecessary trust. The second term charges d_i whenever an action that is legitimately required ($\ell_i = 1$) is blocked ($x_i = 0$): this is the price of operational self-harm. The three constraints then define the feasible region: every legitimate action must be granted (Constraint A.5); no action may be granted unless the verification evidence is sufficient for that resource (Constraint A.6); and each decision is binary (Constraint A.7). In plain terms: permit the minimum standing trust required to keep the business running, provided the evidence clears the bar, and penalise deviations from that minimum in proportion to the damage each type of deviation causes.

The first term of Equation (A.4) thus penalizes unnecessary trust: permitting actions that have no business justification ($\ell_i = 0$), weighted by their risk exposure c_i . The second term penalizes operational self-harm: denying actions that are legitimately required ($\ell_i = 1$), weighted by their denial cost d_i . Constraint (A.5) encodes the business completeness requirement: every action in $\mathcal{L}$ must be permitted. Constraint (A.6) encodes the continuous verification requirement: an action can be permitted only if its verification score $V(a_i)$ meets or exceeds the resource-specific threshold $\theta(\rho_i)$. The verification score $V(a_i)$ aggregates identity assurance level, device trust tier, session freshness, and behavioral conformance into a composite signal; $\theta(\rho_i)$ is a policy-defined minimum that scales with the sensitivity of the target resource ρ_i .

This formulation makes explicit the tradeoff that the original uniform version obscured: the two objective terms pull in opposite directions, and the optimal $\mathbf{x}^*$ is the Pareto point that minimizes their weighted sum given the risk tolerance encoded in c_i and d_i . When $d_i \rightarrow \infty$ for all i , the problem degenerates to full permissiveness (traditional perimeter model). When $c_i \rightarrow \infty$ for all i , it degenerates to full denial (operationally self-defeating deny-all). MST occupies the feasible interior.

Limitations of the formulation. Several limitations of Equation (A.4) deserve explicit acknowledgement.

Action independence. The formulation treats each action a_i as independent: the cost of permitting or denying a_i does not depend on which other actions are simultaneously permitted. In practice, actions are not independent. Permitting a read on a regulated dataset and permitting an export to an external endpoint together create an exfiltration path that neither action creates alone. The independent-action assumption is the most consequential modelling simplification in the formulation, and extending the model to capture combinatorial action interactions—at the cost of substantially increased complexity—is an important direction for future work.

Uniform action weight. Each action is assigned a single scalar c_i , treating all consequences of incorrectly permitting it as commensurable. In practice, an action can have multiple consequence dimensions (regulatory, financial, reputational, operational) that may not reduce to a single number without arbitrary weighting choices. The c_i parameterisation is therefore an abstraction that organisations must calibrate carefully, and different calibrations will produce materially different optimal policies.

The proxy hypothesis. Equation (A.4) optimises a measurable cost function as a proxy for unmeasurable quantities such as $P(\text{breach})$ or expected breach impact. The hypothesis that minimising the weighted trust surplus reduces actual breach probability is operationally motivated but not proven. It is the foundational modelling assumption of the MST framework, not a derived result, and its empirical validation is identified in Section 10 as a priority research direction.

Static $\mathcal{L}$. The legitimate business-required set $\mathcal{L}$ is treated as known and fixed within each optimisation instance. In practice, $\mathcal{L}$ is dynamic, incompletely specified, and contested: engineering teams add new access requirements faster than governance processes can document them, and emergency exceptions create persistent paths that were never formally added to $\mathcal{L}$. The action-space-time graph formulation in Section A.4 partially addresses this by indexing $\mathcal{L}^{(t)}$ over time, but it does not resolve the harder problem of learning $\mathcal{L}$ from operational behaviour rather than from policy declarations.

Action-Space-Time Graph Formulation

The weighted ILP formulation of Equations (A.4)–(A.7) treats the action space as a static flat set. In practice, enterprise access patterns are dynamic: identities, devices, sessions, and resources change state continuously, and the verification score $V(a_i, t)$ and threshold $\theta(\rho_i, t)$ are time-varying quantities. A graph-theoretic formulation captures this temporal dimension naturally.

Define the **action-space-time graph** $G = (V_G, E_G)$ as follows. The vertex set $V_G = \mathcal{U} \cup \mathcal{R}$ partitions into *actor nodes* $\mathcal{U}$ (users, service accounts, processes, devices) and *resource nodes* $\mathcal{R}$ (data stores, APIs, administrative consoles, cloud control plane endpoints). Each directed edge $e_{ij} \in E_G$ from actor node $u_i \in \mathcal{U}$ to resource node $r_j \in \mathcal{R}$ represents a potential access action, with weight c_{ij} encoding risk exposure. Time-indexed copies of the graph, $G^{(t)} = (V_G, E_G^{(t)})$, capture the fact that verification scores and business requirements evolve: an edge e_{ij} that is permitted at time t may not satisfy the verification threshold at time $t + \Delta t$ if device posture degrades or session context changes.

The MST problem on the graph is: find the minimum-cost subgraph $G^* = (V_G, E^*)$ of $G^{(t)}$ such that:

1. E^* contains all edges corresponding to $\mathcal{L}^{(t)}$ (the current legitimate business-required actions);
2. For every edge $e_{ij} \in E^*$: $V(e_{ij}, t) \geq \theta(r_j, t)$;
3. The induced subgraph on $\mathcal{L}^{(t)}$ is connected in the sense that all business workflows can execute without interruption.

The objective function on the graph is $\min_{E^*} \sum_{e_{ij} \in E^* \setminus \mathcal{L}^{(t)}} c_{ij}$, i.e., minimize the total risk-exposure weight of permitted edges that are not legitimately required. This is a *weighted minimum subgraph cover* problem on a bipartite directed graph. When $c_{ij} = 1$ for all edges, it reduces to the uniform cardinality formulation in Equation (A.3).

This graph representation connects directly to intrusion detection systems that model enterprise activity as a provenance graph or process-lineage graph Hassan et al. (2020). In those systems, nodes represent processes, files, and network sockets, and edges represent system-call-level events—precisely the fine granularity of the action space described here. The MST subgraph cover problem can therefore be instantiated on a provenance graph, treating each system-level event as a candidate x_i and applying the verification and risk constraints per edge. This connection makes the MST formulation

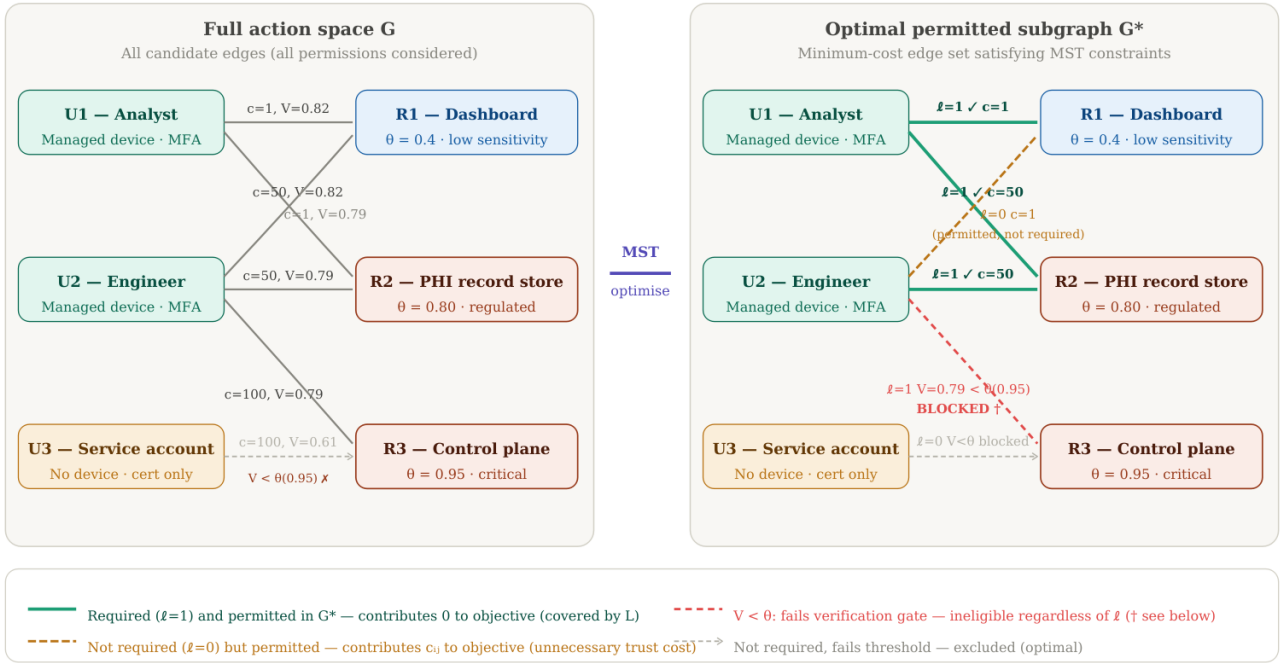


Figure A.1 Weighted minimum subgraph cover: full action space G and optimal permitted subgraph G^* . The left panel shows the full bipartite graph of candidate access actions between three actor nodes ($\mathcal{U}$) and three resource nodes ($\mathcal{R}$), with each directed edge labelled by its risk-exposure weight c_{ij} and current verification score V . The right panel shows the MST-optimal permitted subgraph G^* : green edges are business-required and pass the verification threshold; the amber dashed edge (U2→R1) is not required but has low $c = 1$ cost so is retained; red dashed edges fail the verification gate ($V < \theta$) and are ineligible regardless of business requirement. The key illustrative case is U2→R3: the Engineer needs control-plane access ($\ell = 1$) but achieves only $V = 0.79$, which falls below the critical-resource threshold $\theta = 0.95$. The policy response is not to lower θ but to remediate the evidence gap—step-up authentication or device re-enrolment—before the permission can be granted.

amenable to empirical validation on provenance graph datasets, a direction identified as a priority in Section 10.

Complexity and Tractability

The uniform-cardinality version of the MST problem (minimize $|\mathcal{T} \setminus \mathcal{L}|$ subject to $\mathcal{L} \subseteq \mathcal{T}$ and verification constraints) is equivalent, when the verification constraint is dropped, to finding the minimum superset of $\mathcal{L}$ in $\mathcal{S}$, which is trivially solvable. The computational difficulty arises from two sources: (1) the verification constraint $V(a_i) \geq \theta(p_i)$ couples the permission variables x_i to a scoring function that may be non-linear and resource-intensive to evaluate; (2) in the weighted formulation, the interaction between the two objective terms and the completeness constraint creates a structure analogous to the **0-1 knapsack problem**, which is NP-hard in the strong sense Garey and Johnson (1979): each action a_i has a “profit” (denial cost d_i if it is in $\mathcal{L}$) and a “weight” (risk cost c_i), and the problem is to select the minimum-risk superset of $\mathcal{L}$ within the feasible verification region.

More precisely, the weighted MST problem (Equations A.4–A.7) is NP-hard by reduction from 0-1 knapsack: given a knapsack instance

with items $\{1, \dots, n\}$, values v_i , weights w_i , and capacity W , construct an MST instance with $\mathcal{L} = \emptyset$, $c_i = w_i$, $d_i = v_i$, and $\theta(p_i) = 0$ for all i . The MST optimal selects the minimum-weight feasible set, which corresponds directly to the knapsack optimal selection.

For practical enterprise scales, this complexity motivates relaxation approaches. A **linear programming (LP) relaxation** of Equations (A.4)–(A.7) replaces $x_i \in \{0, 1\}$ with $x_i \in [0, 1]$ and solves the resulting LP in polynomial time; the fractional solution can be rounded to a feasible integer solution with a bounded approximation ratio. Alternatively, the structure of the problem—bipartite graph, separable objective, binary variables—makes it well-suited to **branch-and-bound mixed integer programming (MIP)** solvers, which handle enterprise-scale instances (tens of thousands of action types) routinely in practice Wolsey (1998). For real-time enforcement decisions, where the full MIP is too slow, a greedy policy that sorts candidate actions by $c_i/V(a_i, t)$ (risk per unit of verification evidence) and admits actions in ascending risk

order until the completeness constraint is satisfied provides a practical approximation with a well-understood performance bound.

The time-varying graph formulation introduces additional structure: as $\theta(r_j, t)$ and $V(e_{ij}, t)$ change, previously permitted edges may need to be revoked and previously denied edges may become admissible. The dynamic version of the problem is related to *online subgraph maintenance* under changing constraints, a problem studied in the streaming graph algorithms literature McGregor (2014). This connection suggests that the SOC's continuous re-evaluation of session trust is not merely a policy preference but a computational necessity: the optimal permitted set changes as the verification evidence changes, and maintaining it requires incremental update rather than full re-solve.

Mapping Optimization Elements to Architecture

The formalization above maps directly onto the two-plane operating architecture and unified telemetry framework described in Sections 4–7, as summarized below.

Action space $\mathcal{S}$. At the external application edge, the action space is narrow: inbound HTTP/S requests, API calls, authentication attempts, and partner integrations. The cardinality of $\mathcal{L}$ at the edge is small and well-documented, making the optimization tractable and the verification threshold θ enforceable close to deny-by-default. Internally, $|\mathcal{S}|$ is orders of magnitude larger (every user-to-resource path across the enterprise), but the risk weights c_i concentrate on a small subset of high-consequence actions (privileged administrative sessions, regulated data access, cloud control plane operations), allowing the optimizer to allocate verification resources proportionally.

Verification score $V(a_i, t)$. The unified telemetry plane (Section 7) exists precisely to compute $V(a_i, t)$ in real time. Each source—IAM, EDR, ZTNA, NIDS, UEBA—contributes a component of the verification score. Without telemetry fusion, $V(a_i, t)$ cannot be computed accurately, and the optimization problem becomes infeasible: actions are permitted or denied on incomplete evidence, which corresponds to setting some c_i to zero (under-penalizing risky actions) or some d_i to zero (under-protecting legitimate ones).

Bounded autonomy as a constraint on c_i . The bounded-autonomy ladder (Section 8) is a policy translation of the risk-weight structure. Actions at the top of the ladder (policy rewrites, permanent access changes) have high c_i and are therefore excluded from autonomous execution even when $V(a_i, t) \geq \theta$. Actions at the bottom (alert enrichment, duplicate suppression) have negligible c_i and are freely automated. The ladder makes the optimization structure operationally legible to analysts who do not interact with the formal model directly.

Discussion of assumptions. The MST model rests on the working hypothesis that minimizing the risk-weighted unnecessary trust set $\sum_i c_i x_i (1 - \ell_i)$ is a useful proxy for minimizing concrete security metrics such as $P(\text{breach})$ or expected breach impact. This hypothesis is not proven in the paper; it is the fundamental modeling assumption, analogous to the assumption in network hardening that minimizing attack graph reach reduces breach probability Shandilya and Nagar (2014). The advantage of the set-size and graph-size formulations is that they are computable from operational data, whereas $P(\text{breach})$ is not. We acknowledge this approximation and identify its empirical validation as a priority research direction (Section 10).

Attack Surface Minimization as the Theoretical Objective

The formulation of Equations (A.3)–(A.7) has a direct interpretation in the attack-surface-minimization literature introduced by work on attack surface metrics Howard et al. (2005); Manadhata and Wing (2011). In that tradition, the *attack surface* of a system is the subset of its resources reachable and usable by an attacker: entry points through which inputs flow, exit points through which outputs flow, and channel resources that mediate communication. Reducing this surface reduces the opportunities available for adversarial exploitation.

For the MST framework, the connection is immediate. The permitted action set $\mathcal{T} \subseteq \mathcal{S}$ is the enterprise attack surface in the relevant sense: every action permitted by policy is an action an attacker might exploit if they can impersonate a legitimate actor or compromise the verification context. The legitimate business-required set $\mathcal{L} \subseteq \mathcal{T}$ is the irreducible lower bound on this surface—the portion that must be exposed for the service to function—and the trust surplus $\mathcal{T} \setminus \mathcal{L}$ is the avoidable portion. The MST objective is therefore a direct quantitative expression of attack surface minimization under operational feasibility constraints.

The effect of reputation-based blocking. A reputation-list control—whether operated at the network perimeter (NIDS, firewall) or at the endpoint (EDR egress policy)—modifies the problem structure by pre-filtering the action space itself, not merely the permitted subset. Let $\mathcal{B} \subseteq \mathcal{S}$ denote the set of actions whose identity tuple $a_i = (\iota_i, \delta_i, \rho_i, \sigma_i, \omega_i)$ has either ι_i or ρ_i resolving to an IP address on a curated reputation feed. Reputation-based auto-blocking enforces $x_i = 0$ for every $a_i \in \mathcal{B}$ regardless of all other considerations. The effective action space becomes:

$$\mathcal{S}_{\text{eff}} = \mathcal{S} \setminus \mathcal{B} \quad (\text{A.8})$$

and the MST optimization is solved over $\mathcal{S}_{\text{eff}}$ rather than $\mathcal{S}$. Because reputation feeds explicitly enumerate hosts with documented adversarial activity, the expected intersection $\mathcal{L} \cap \mathcal{B}$ is close to empty—legitimate business counterparties do not appear on Spamhaus DROP or AbuseIPDB—so the completeness constraint $\mathcal{L} \subseteq \mathcal{T}^*$ remains satisfiable in $\mathcal{S}_{\text{eff}}$. The practical result is that the optimal permitted set $\mathcal{T}_{\text{eff}}^*$ computed over $\mathcal{S}_{\text{eff}}$ has strictly smaller attacker-exploitable surface than the corresponding optimum $\mathcal{T}^*$ computed over the full $\mathcal{S}$:

$$|\mathcal{T}_{\text{eff}}^* \setminus \mathcal{L}| \leq |\mathcal{T}^* \setminus \mathcal{L}| \quad (\text{A.9})$$

with strict inequality whenever $\mathcal{B}$ contains any action that would otherwise satisfy the verification threshold under $\mathcal{T}^*$.

The theoretical lower bound. Equation (A.9) frames the fundamental question of how far attack surface can be reduced in principle. The global lower bound is given by the legitimate set itself:

$$|\mathcal{T}^* \setminus \mathcal{L}| \geq 0 \quad (\text{A.10})$$

with equality if and only if every action in $\mathcal{T}^*$ is business-required. This bound is achievable only under perfect information—complete knowledge of $\mathcal{L}$, perfect verification ($V(a_i) = 1$ for all legitimate actions), and no external dependencies. In practice, the realized trust surplus is strictly positive, and the engineering question becomes: how close to zero can it be driven without violating the operational-cost constraint?

The two planes of the MST architecture give different answers to this question. At the external application edge, $\mathcal{L}$ is narrow, well-documented, and relatively static (customer API calls, authentication endpoints, partner integrations), making the trust surplus easier to drive toward zero. Bidirectional reputation blocking reduces it further by shrinking the action space before the optimization is even formulated. Inside the internal enterprise, $\mathcal{L}$ is broader, dynamic, and incompletely specified, so the achievable minimum is inherently larger—but endpoint-enforced reputation blocking still reduces it by an amount bounded by the size of $\mathcal{B}$ restricted to internal-origin traffic.

The MST model thus provides both an operational framework and a theoretical minimum: the ideal end-state of a Zero Trust program

is a permitted action set exactly equal to the current legitimate business-required set, with all other actions either denied by policy or pre-filtered by reputation-based blocking. Every deviation from this ideal—every permitted action outside $\mathcal{L}$, every connection to or from a reputation-listed IP that the platform fails to block—is a quantifiable gap in the architecture rather than an unmeasurable policy judgement. This quantifiability is what distinguishes attack surface minimization from traditional Zero Trust rhetoric: it supplies a target that organizations can measure progress against, and it supplies a theoretical justification for specific controls (reputation blocking, JIT privilege, device tiering) in terms of the surface-reduction they produce.